OPD Saathi

SMART INDIA HACKATHON · SIH26047

Complete product and technical documentation

A multilingual check-in kiosk, phone check-in and live queue for crowded government OPDs. The patient's history is ready before they reach the doctor.

ओपीडी साथी

Hindi · Marathi · English | Tap, speak or type | Explainable triage | Doctor-reviewed AI drafts | Encrypted, per-hospital data | History only, no diagnosis

VERSION
main @ cd5f9e5
Phases 0–6 complete

DATE
17 October 2026

DATA
All patients, staff and hospitals shown are fictional

About this document

This is the single reference for OPD Saathi: why it exists, what every screen does, how every rule and algorithm works, how the system is built, how it is secured, tested and run, and what remains before a real hospital can use it.

Who it is for

- **Team members and new contributors** who need to understand the whole system quickly.
- **Reviewers and evaluators** who want to know exactly what is built and how it behaves.
- **Hospital staff and IT** who will set it up and run it (Part III).

How to read it

- **Part I** describes the product screen by screen, for everyone.
- **Part II** explains the inside: architecture, algorithms with their exact numbers, data, API, security, quality.
- **Part III** covers running it, troubleshooting, limitations and the road to production.

Conventions

Marker	Meaning
`services/api/app/...`	A file in the repository; paths are relative to the repository root.
P0 P1 P2	Priorities used by the production plan: P0 before any real patient, P1 before going beyond a pilot, P2 afterwards.
Callout boxes	Green: how something works. Amber: a caveat. Red: a safety rule. Violet: AI behaviour.
Numbers	Every number in the algorithm chapters (weights, limits, timeouts) is taken from the code on `main` at commit `cd5f9e5`.

Safety framing, in one paragraph

OPD Saathi collects a patient's history and helps staff run a fair queue. It never diagnoses, names a likely disease, suggests medicines or doses, gives treatment advice or reassures patients. Deterministic rules decide urgency; AI can only add caution, and every AI output is a labelled draft that a person accepts or rejects. All clinical content (questions, red-flag rules, triage weights, Ayurveda items, consent text) is a draft awaiting clinician and native-speaker review. Only synthetic data has ever been used.

Where the other documents are

This document consolidates the repository's documentation. The source documents stay authoritative for their topics:

Document	Topic
`docs/PHASE_7_PRODUCTION_PLAN.md`	The final phase: everything left before a real hospital deployment (131 tasks, gates, go-live checklist).
`docs/IITB_READINESS_PLAN.md`	Audit, requirements matrix against SIH26047, and the Phase 0–6 build log with test evidence.
`docs/QUEUE_POLICY.md`	Calling order, missed calls, recall, permissions, decisions the hospital signs off.
`docs/PRIVACY.md`	Data map, consent by purpose, old papers, returning patients, retention.
`docs/E2E_SCENARIOS.md`	The fifteen end-to-end scenarios, their tests and the defects they found.
`docs/EVALUATION.md`	Synthetic evaluation results for red flags and routing.
`docs/DEMO.md`, `docs/MOBILE_LAN.md`, `docs/DEPLOYMENT.md`	Demo script, phones on the hospital Wi-Fi, installing at a hospital.

Document	Topic
`docs/DECISIONS.md`, `docs/SAFETY.md`, `docs/CLINICIAN_REVIEW.md`	Architecture decisions, safety framing, the clinical review queue.
`docs/SCALING.md`, `docs/OBSERVABILITY.md`	Capacity notes and the optional Prometheus/Grafana profile.

Contents

PART I · THE PRODUCT

1 Why OPD Saathi exists

2 The product at a glance

3 The patient kiosk, step by step

Welcome · consent · AI consent · details · OPD card · safety check · interview · review and old papers · token · comfort and help · offline form · kiosk activation

4 Patients' own phones

Phone check-in QR · live status page · missed call · Stop AI help

5 The staff workspace

6 Queue dashboard (nurses and doctors)

7 The doctor's consultation workspace

Overview and AI draft · history · previous visits · old papers · Ayurveda · transcript · notes · timeline · PDF

8 Patients register and assisted registration

9 Reports

10 Settings and administration

11 AI in OPD Saathi

12 One system, many hospitals

PART II · HOW IT WORKS INSIDE

13 Architecture

14 Technology stack

15 Repository map

16 Algorithms and rules

Interview engine · red flags · triage score · calling order · wait estimate · routing · queue state machine · tokens and codes · AI pipeline · paper flags · notifications · reports · limits and idempotency · day close-out

17 Data model

18 API

19 Security and privacy

20 Offline use and the PWA

21 Quality: tests, CI and evaluations

PART III · RUNNING IT

22 Running the demo

23 Installing at a hospital

24 Day-to-day operations and commands

25 Troubleshooting

26 Limitations and the road to production

27 Glossary

APPENDICES

A All 96 API routes

B The question bank

PART I

The product

What OPD Saathi is for, who uses it, and what every screen does, from the patient's first tap at the kiosk to the administrator's reports.

1. Why OPD Saathi exists
2. The product at a glance
3. The patient kiosk, step by step
4. Patients' own phones
5. The staff workspace
6. Queue dashboard
7. The doctor's consultation workspace
8. Patients register
9. Reports
10. Settings and administration
11. AI in OPD Saathi
12. One system, many hospitals

1 Why OPD Saathi exists

Government outpatient departments (OPDs) see hundreds of patients every morning. Consultations last a few minutes, patients speak many languages and describe their problems in their own way, papers from earlier visits are scattered, and nobody in the waiting hall knows when their turn will come. The doctor spends much of those few minutes collecting basic history.

The problem statement: SIH26047

Title	Patient Case-Taking Software (SIH26047)
Organisation	Ministry of Ayush · All India Institute of Ayurveda (AIIA)
Category · theme	Software · MedTech / BioTech / HealthTech
The need	Histories are incomplete because consultations are short; AYUSH practice needs an even more extensive history (Prakriti, Vikriti, Agni); patients bring fragmented paper records; ABDM exists, but there is no patient-facing "first mile" tool that captures history and digitises documents before the visit.
Expected journey	Identify and consent → converse → scan documents → summarise and route → physician reviews.

The wording was verified from two independent copies of the official entry (the official portal could not be fetched automatically). See `docs/PROBLEM_STATEMENT.md`.

The four modules asked for, and what OPD Saathi delivers

Module	What is built	What is not built yet (and why)
A. Conversational history engine	An adaptive interview by voice, touch or typing in Hindi, Marathi and English (chapter 3, section 16.1); complaint-specific question modules; up to three AI-written follow-up questions, validated like bank questions; surgical and personal history; "don't know" and "prefer not to say" kept apart from "no"; an optional Ayurveda history; danger-sign tiles and 15 red-flag rules that raise urgent staff alerts (16.2).	A short review-of-systems screen (Phase 7 task 7.11.2). All clinical content still needs clinician sign-off.
B. Document digitisation	The patient photographs old prescriptions, test reports and discharge papers; with their consent AI reads each into an unconfirmed draft (diagnoses, medicines, results with the printed reference range, procedures, the printed date); the doctor compares draft and photo and accepts or rejects it; results are flagged above/below/within only against the range printed on the same paper (16.10).	A merged timeline across papers (7.11.3); drug-interaction checks and general reference ranges are deliberately excluded until a licensed, reviewed knowledge base exists.
C. Summary generator	A structured, versioned English case sheet (complaint, history of present illness, past history and operations, medicines, allergies, family and personal history, open questions); an AI brief in which every sentence cites the answers it rests on, which the doctor accepts, rewrites or rejects; the patient hears and confirms a read-back in their own language; a printable PDF (chapter 7).	Review of systems and a "prior investigations" section built from accepted readings (7.11.2, 7.11.3). The doctor's own examination and plan notes (7.11.1).

Module	What is built	What is not built yet (and why)
D. Consent, privacy and ABDM	Consent by purpose: care (required) and online AI help (optional, withdrawable from the patient's own phone); encrypted patient fields, transcripts, case sheets and photos; per-hospital isolation inside the database; session data cleared after submission; an append-only audit log (chapter 19).	ABHA authentication, ABDM linking and FHIR transfer (7.10); a legal DPDP review (7.5).

What OPD Saathi deliberately does not do

Never

- diagnose, name a likely disease, suggest medicines or doses, or give treatment advice;
- show a patient their urgency score, or anyone else's position;
- let AI lower urgency, or decide urgency on its own;
- link two patient records automatically from a phone number or name.

Always

- ask for consent before collecting anything;
- let the patient hear and confirm their own history;
- show staff why a patient is where they are in the queue;
- keep working without internet and without AI (rules, tapping and typing).

Design principles

Principle	How it shows up
History, never diagnosis	Neutral patient wording; an output guard rejects AI text that diagnoses, advises or prescribes; the case sheet schema has no diagnosis field.
Rules decide, AI checks, people confirm	Danger tiles and red-flag rules set urgency; an AI "urgent" counts as priority at most until a nurse confirms; AI drafts are reviewed by a doctor.
Uncertainty escalates	Urgency never drops without a recorded staff decision; unclear routing goes to nurse triage; a self-reported danger sign stays "Nurse to confirm".
Explainable	Every triage point is listed with its reason; every AI sentence cites its answers; every queue change has a timeline entry and an audit record.
Works on a bad day	No internet: rules, tapping, typing and the local queue keep working; flaky Wi-Fi: answers and confirmations are safe to retry; no network at the kiosk: an encrypted short form.
Private by default	Only age and sex are required; AI is opt-in per patient; one installation per hospital; patient data encrypted at rest; tenant isolation enforced by the database.

2 The product at a glance

One web application serves five audiences. Patients never need an account; staff screens change with the signed-in role.

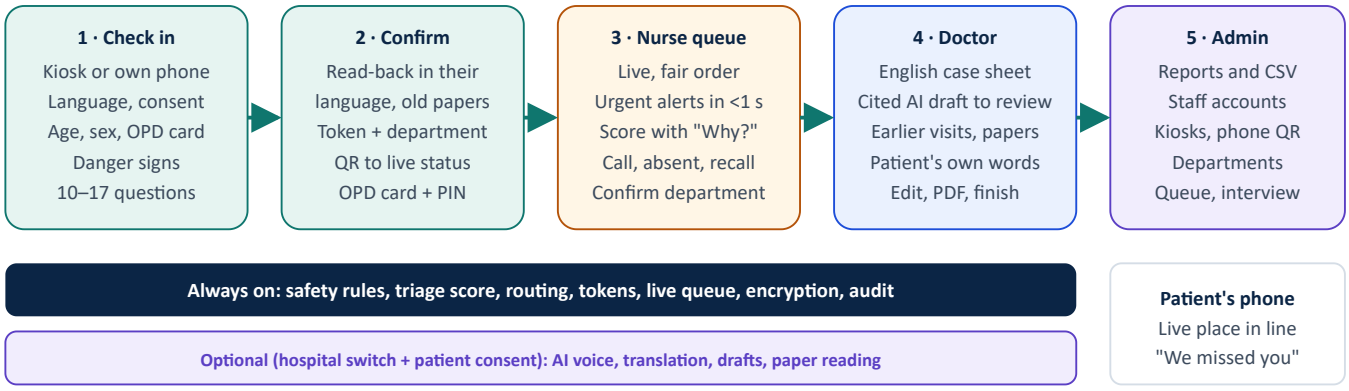


Figure 2.1 One patient's journey through OPD Saathi.

Who uses what

Who	Where	What they do there
Patient at a kiosk	`/kiosk` → `/kiosk/intake`	Chooses Hindi, Marathi or English; agrees to the care notice and answers the separate AI question; gives age and sex (phone optional); links today's visit with their OPD card if they have one; flags danger signs; answers 10–17 questions by tapping, speaking or typing; can photograph old papers; checks a read-back; gets a token, department, place in line, a QR code and an OPD card number with PIN.
Patient's own phone	`/m/<code>` (check-in QR), `/s/<token>` (status)	Checks in from a QR poster with the same interview, or follows their turn live from the token's QR: place in line, estimated wait, progress, "we missed you", and a button to stop AI help.
Nurse / reception	Staff → Queue, Patients	Watches the live queue and alerts, calls patients (without reserving them for a doctor), marks absent and recalls, changes priority or department with a reason, confirms uncertain departments, links returning patients at the desk, registers patients who need help.
Doctor	Staff → Queue, Consultations, Patients	Calls the next patient (the server chooses), opens the case sheet, reviews red flags and the cited AI draft, earlier visits and old papers, corrects fields, records Ayurveda findings, starts and finishes the consultation, prints the PDF.
Administrator	All staff pages, plus Reports and Settings	Reads reports and exports CSV; manages staff accounts, kiosks, phone check-in QRs and departments; sets the call grace period and the Ayurveda module; sets the "time saved" assumption.

Screens and addresses

Address	Who	Screen
`/kiosk`, `/kiosk?activate=<code>`	Kiosk device	Welcome and language; first-time activation of a kiosk with a one-time code
`/kiosk/intake`	Patient	The check-in flow: consent → details → safety check → questions → review → token
`/kiosk/offline-intake`	Patient	Encrypted short form when the kiosk has no network

Address	Who	Screen
`/m/<code>`, `/m/<code>/intake`	Patient's phone	Phone check-in landing page and the same interview on the phone
`/s/<token>`	Patient's phone	Live queue status for one visit
`/staff` → `/staff/queue`	Nurse, doctor, admin	Sign-in, then the live queue dashboard
`/staff/consultations`	Doctor, nurse, admin	Case sheet workspace with tabs
`/staff/patients`	Nurse, doctor, admin	Visits register and assisted registration
`/staff/reports`, `/staff/settings`	Admin	Reports and CSV; users, kiosks, departments and hospital settings

In numbers

3
languages everywhere (1,134 interface strings each, parity-checked)

17
questions at most (+6 with the Ayurveda module)

15 · 6
red-flag rules · danger-sign tiles

8
standard departments with automatic routing

96
API routes, each with server-side authorisation

22
database migrations, all reversible

965
API tests passing (core and domain coverage 96%)

96 · 144
web unit tests · browser (Playwright) tests

15/15
end-to-end scenarios covered by tests

3 The patient kiosk, step by step

A touch-screen kiosk: any tablet or PC with Chrome or Edge. Six steps are shown at the top of every screen: **Agree → Details → Safety check → Questions → Review → Token**. The patient can change language, go back, call staff or stop at any point.

3.1 Welcome and language

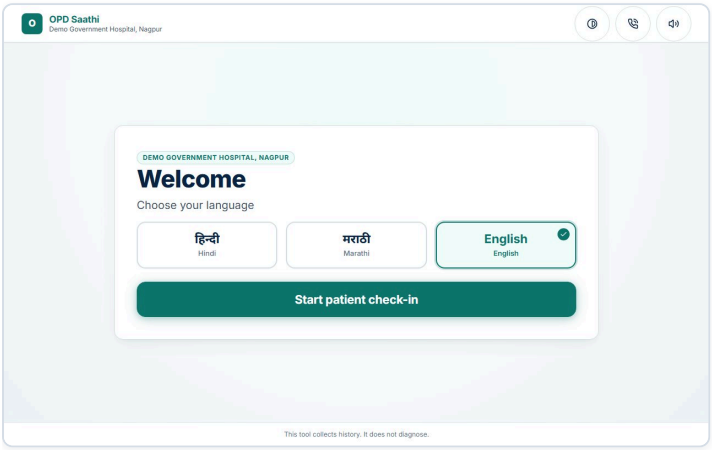


Figure 3.1 Welcome. The three round buttons switch voice guidance, high contrast and "Call staff".

The patient taps हिन्दी, मराठी or English and **Start patient check-in**. The kiosk starts a server session (`POST /intake/sessions`), which creates a visit in the `registering` state and returns a two-hour patient token and the consent notice in that language.

Everything after this point happens in the chosen language and can be switched mid-way; earlier answers are kept as given. The hospital's name is shown at the top of every kiosk screen.

3.2 Agree: the care notice

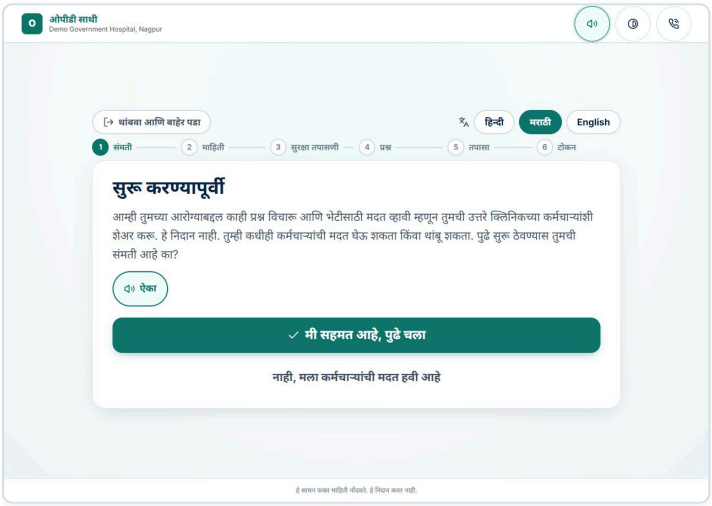


Figure 3.2 The care notice in Marathi, read aloud automatically; "Listen" repeats it.

- Nothing is collected before the patient agrees. The notice explains that the answers go to the clinic team and that this is not a diagnosis.
- Each notice has a version and a SHA-256 hash of its text; the server stores the answer with the version, hash, language, channel (kiosk, phone or nurse) and time, so a "no" is recorded as precisely as a "yes".
- **"No, I need staff help"** ends the check-in and sends the patient to the desk.

3.3 Online AI help: a separate, optional question

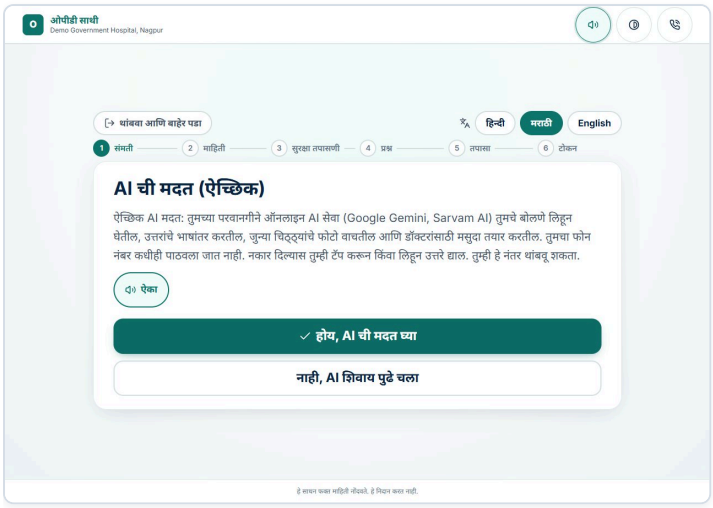


Figure 3.3 The AI question. Both answers are styled alike and nothing is pre-selected.

What "yes" and "no" mean

Yes: online services (Google Gemini, Sarvam AI) may turn the patient's voice into text, translate answers, read photos of old papers and prepare a draft for the doctor. The phone number is never sent.

No: the visit continues exactly the same way on this server alone. The patient taps and types, the read-back uses a voice installed on the device, and the doctor gets the rule-based case sheet with the patient's own words. The API enforces this, not just the screen. A "yes" can be changed until confirmation and withdrawn afterwards (chapter 4).

3.4 Details: age, sex and an optional phone

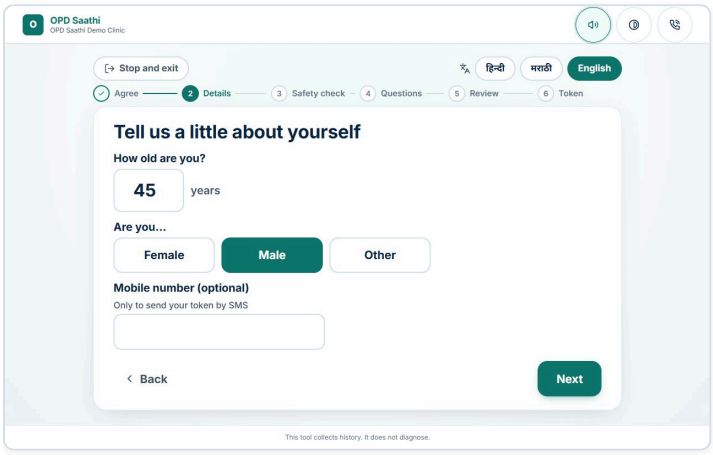


Figure 3.4 Details. Only age and sex are required.

Age and sex shape the interview (no pregnancy questions for men or children) and the triage score. A mobile number is optional and is used only for SMS queue updates if the hospital connects an SMS provider. It is stored encrypted, with a keyed digest for lookup; a phone number never links records by itself, because families share phones.

Known defect, fixed first in Phase 7 (task 7.0.2)

Typing a 10-digit number currently ticks the SMS opt-in box automatically. Consent must be the patient's own tap; the box will be unticked by default and SMS will become its own recorded consent purpose.

3.5 Returning patients: the OPD card

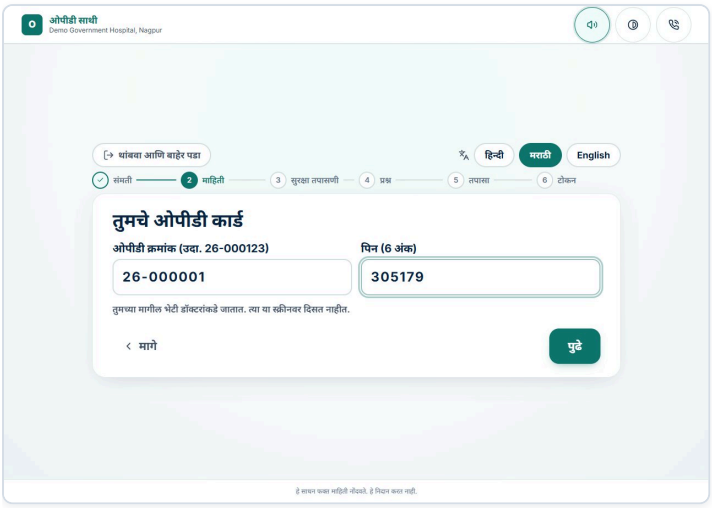


Figure 3.5 A returning patient enters the OPD number and PIN from their last token slip (synthetic card).

After age and sex the kiosk asks **"Have you visited this hospital before?"**. "Yes" opens this card step; "No" goes straight to the safety check. A nurse-assisted check-in skips it, because the nurse links records at the desk.

If the card links, the kiosk shows only the date of the last visit and asks whether today continues that visit (a follow-up) or is a new problem. The rules behind the card are below.

- Every patient gets an **OPD number** (format `26-000123`: year and a per-hospital sequence) when their first visit is confirmed, and a **six-digit PIN**. The PIN is derived with the hospital's HMAC key and is never stored (section 16.8).
- Entering both links today's visit to the existing record, provided today's age and sex fit it (birth year within one year; female/male must agree). A wrong number, wrong PIN or mismatch is refused without saying which part failed. Limits: 5 attempts per visit in 15 minutes, 10 per OPD number per hour. Every outcome is audited (without the number or PIN).
- The shared kiosk never shows what an earlier visit was about: privacy on a public screen.
- Without the card, a nurse can search and link at the desk (chapter 6). Records are never merged automatically.

3.6 Safety check: danger signs first

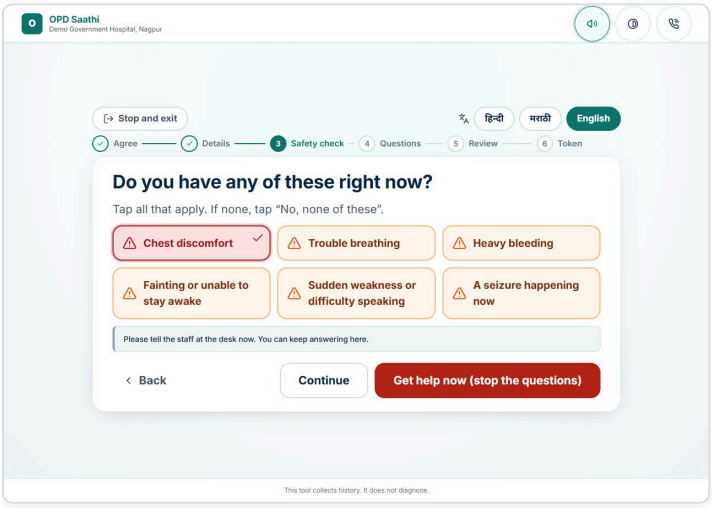


Figure 3.6 Six danger-sign tiles come before any question; "Chest discomfort" selected.

- Tiles: chest discomfort, trouble breathing, heavy bleeding, fainting or unable to stay awake, sudden weakness or difficulty speaking, a seizure happening now.
- Tapping a tile makes the visit **urgent at once** and raises an alert on every staff board of the department within a second (measured p95 under 1 s). The patient's row stays marked **Nurse to confirm** until a nurse acknowledges it after seeing them.
- **Get help now (stop the questions)** opens an emergency screen telling the patient to go to the staff now; **Continue** lets them keep answering while staff come.
- There is no anonymous "emergency" button: it was removed because anyone could raise false alarms. Real emergencies are handled by staff in person.

3.7 Questions: tap, speak or type

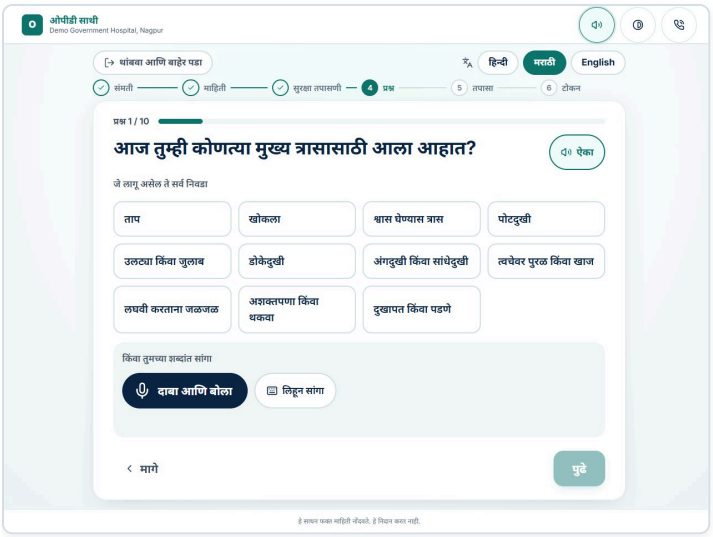


Figure 3.7 "What is the main problem?" with tap-to-answer tiles, **Press and speak** and **Type instead**.

- **One question at a time**, with a progress bar ("Question 1 / 10"). Tiles can be combined with the patient's own words.
- **The interview adapts.** The main problem selects up to two symptom modules (fever, breathing, stomach, urine, pain, skin, injury, sleep and energy); then come past illnesses, operations, pregnancy (only where it can apply), medicines, allergies, family history, tobacco and alcohol, and "anything else". The full algorithm is in section 16.1.

- **AI follow-ups** (with AI consent): up to three extra questions written for this complaint, in all three languages with tap options, checked like bank questions and labelled "AI follow-up" for the doctor.
- **Voice:** the question is read aloud ("Listen" repeats it); a spoken answer is transcribed (Sarvam, then Gemini, then the browser's own recogniser) and shown for checking before it is used. Voice needs a secure page (HTTPS or the kiosk PC itself); otherwise the screen says so and offers tap and type.
- **Honest answers:** "Don't know" and "Prefer not to say" are recorded as open questions, never as "no".
- **Prescription scan:** on the medicines question the patient can photograph an old prescription; with AI consent the medicine names are read and shown back for confirmation. The photo is processed in memory and never stored.
- **Back** changes the previous answer (it is shown pre-filled); changing the main problem re-plans the symptom questions. **Stop and exit** ends the check-in safely: a visit without a token is cancelled and never reaches the queue.
- Every answer is saved at once as an encrypted interview turn, and re-runs the red-flag rules and the triage score. An answer sent twice (a double tap, a retry after lost Wi-Fi) is saved once; the kiosk says "tap again, your answers are kept".

3.8 Review: read-back and old papers

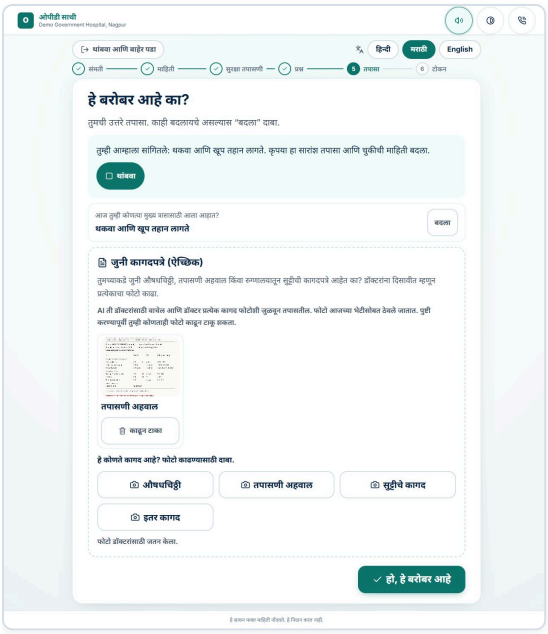


Figure 3.8 The review in Marathi: the read-back, **Change** on each answer, and one test report added under old papers.

- The whole history is read back in the patient's language. The read-back text is built from fixed templates, never by a language model.
- **Old papers (optional):** the patient first taps what the paper is (prescription, test report, discharge paper, other), then photographs it. Up to five papers; each can be removed until confirmation. Only real JPEG, PNG or WebP photos up to 1 MB are kept, encrypted. Details in sections 7.4 and 16.10.
- **Yes, this is correct** confirms. From then on the answers cannot be changed by the patient; staff can still correct the case sheet, with every edit versioned and audited.
- If a patient stays on the review screen for 90 seconds without confirming, a nurse is asked to help.

3.9 Token, QR and OPD card

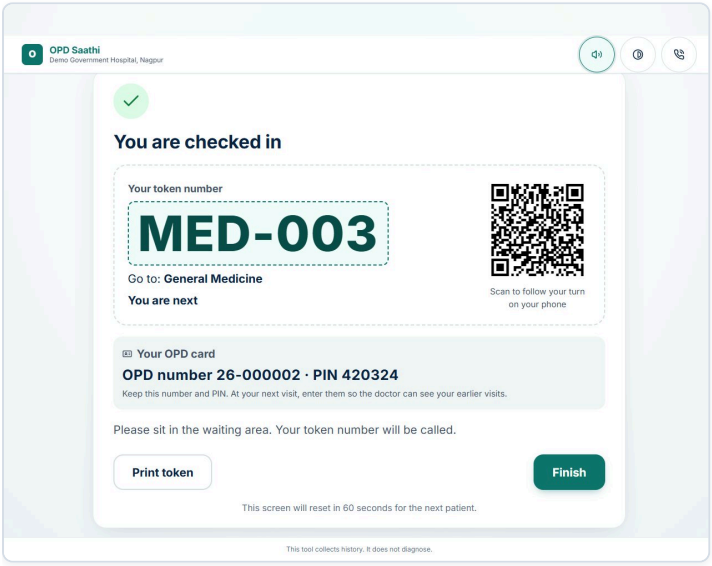


Figure 3.9 Token, department, place in line, the QR code for the phone and the OPD card (synthetic values).

Confirmation (`POST /intake/sessions/{id}/confirm`) issues the day's token for the department (for example `MED-003`) exactly once, even if the request is repeated or sent twice at the same moment.

Print token prints a slip; **Finish** or 60 seconds of inactivity resets the kiosk for the next patient and clears everything from the browser. The QR opens the live status page (chapter 4); the OPD number and PIN let the patient link their next visit.

3.10 Help, comfort and accessibility

Always within reach

- **Call staff** on every screen raises a "Help requested" alert on the staff board (limited to 5 presses per visit in 10 minutes, and not after the visit has finished).
- **Voice guidance** reads every screen; one tap turns it off.
- **High contrast** mode for low vision.
- Every button is at least 64 px, easy for older hands.

Privacy on a shared screen

- After 60 seconds without a tap the kiosk warns, then resets; the next patient never sees someone else's answers.
- The kiosk keeps its patient session in tab-scoped storage only.
- The shared screen never shows earlier visits' details.
- The kiosk shows neutral wording only: no scores, no disease names.

3.11 When the kiosk has no network: the offline short form

If the kiosk loses its connection (and the page is in a secure context, which Web Crypto needs), the welcome screen offers a short form: complaint, duration, severity, long-term conditions, medicines, allergies and danger signs, with the same consent first. The form is encrypted in the browser's IndexedDB with AES-GCM under a non-extractable key, and the patient gets a provisional receipt. When the connection returns, the kiosk replays each form once to `POST /intake/offline-submissions` with its idempotency key, through the regular intake engine; staff then confirm the real token. Sending twice can never create two visits, and records are deleted from the device after a confirmed sync.

> **Limit**
> The offline form covers a kiosk that has lost its network. If the server itself is down, check-in stops; Phase 7 adds a paper fallback procedure (task 7.7.7).

3.12 Setting up a kiosk

1. An administrator registers the kiosk in Settings → Kiosks and gets a **one-time activation code** valid for 24 hours, with **Copy setup link**.
2. Opening the link once on the kiosk device (`/kiosk?activate=…`) exchanges the code for a long-lived **device secret**; from then on the kiosk signs itself in (12-hour kiosk tokens, renewed with the secret), overnight and after restarts.
3. The kiosk sends a heartbeat about once a minute, so Settings shows it as **Online**, **Offline**, **Not seen yet** or **Turned off**.
4. A replaced device gets a **New activation code** for the same kiosk record; its history is kept and the old device is signed out once the new one pairs. **Deactivate** pauses it; **Delete** cuts it off at once.

4 Patients' own phones

Patients can check in on their own phones from a QR poster, and every patient can follow their turn live from the QR on their token. No app and no login are needed.

4.1 Phone check-in from a QR poster

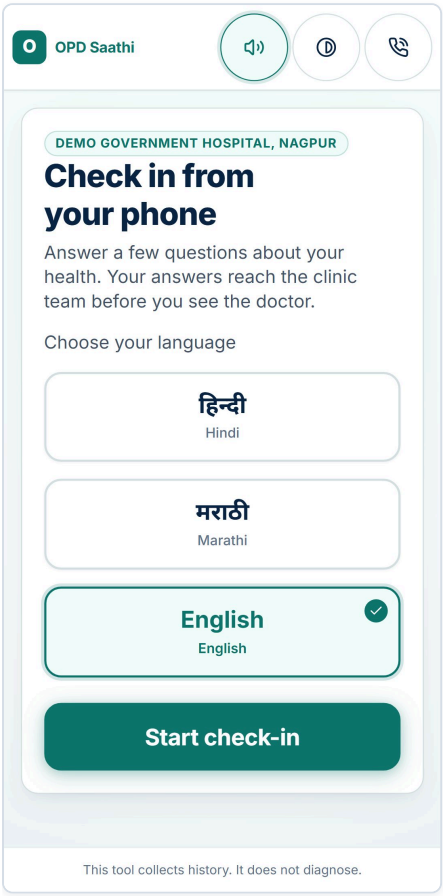


Figure 4.1 The poster's QR opens this page.



Figure 4.2 The same interview, in Hindi, on a phone.

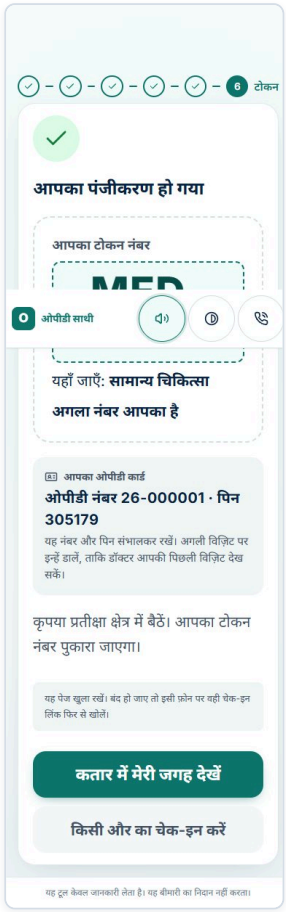


Figure 4.3 The phone keeps its token.

- An administrator creates a **Phone check-in QR** (Settings → Kiosks → Type) and prints it for the waiting hall. The code inside is derived from the hospital key; **Replace QR** makes old posters stop working at once; **Deactivate** pauses phone check-in.
- The patient completes the **same** server-side interview as at a kiosk, with the same consent, safety rules and token. Staff see a **Phone** badge on these patients so the desk can greet them.
- Reloading the page or losing Wi-Fi resumes the same visit at the same question. After the token is issued, reopening the link shows "You already have token MED-012" with a button to the live page and an option to check in someone else.
- Abuse limits: one phone can start at most 6 check-ins per QR in 10 minutes, one QR at most 150 in 10 minutes; patients enter the queue only after they confirm.
- Over plain HTTP on the hospital Wi-Fi the microphone is not available, so the page says "voice needs the hospital's secure link" and offers tapping and typing. HTTPS is a Phase 7 task (7.2.1).

4.2 The live status page

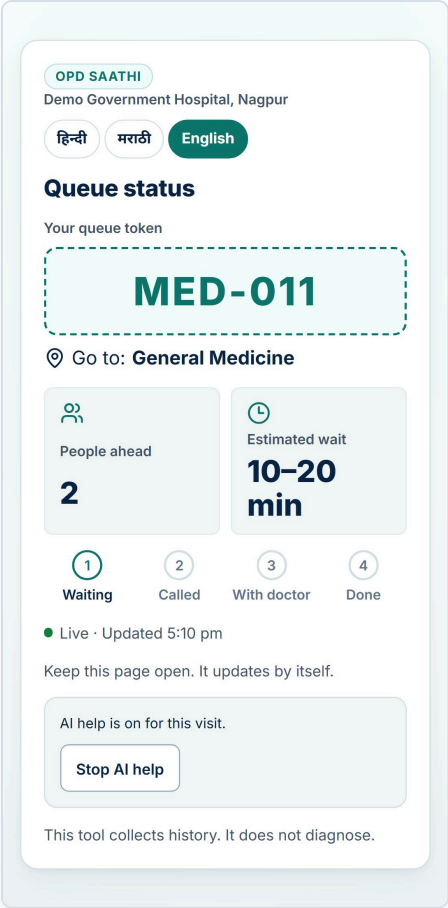


Figure 4.4 Waiting: people ahead, a wait range, progress and **Stop AI help**.



Figure 4.5 A missed call, in Hindi: "please go to the registration desk; staff will put you back with the same token".

- **What it shows:** the token and department, people ahead, an estimated wait as a five-minute range (never a false-precise time, section 16.5), a four-step progress track (Waiting → Called → With doctor → Done), and the language switch. It updates by itself over a server-sent event stream, falling back to polling.
- **When called** the phone vibrates and the tab title changes, so the moment is not missed.
- **Missed call:** if staff mark the patient absent, the page asks them to go to the desk; their token and place are kept for a recall.
- **Stop AI help:** a patient who said yes can withdraw from here. The link can only *stop* AI help, never start it, so a photographed token slip cannot switch it on. From then on nothing more about the visit goes to AI services, the doctor no longer sees the AI draft, and it cannot be reviewed.
- **Privacy:** the address contains a 256-bit opaque token derived from the hospital key and stored only as a hash; the page shows no name or health details.
- **SMS (optional):** if the hospital connects a provider and the patient opts in, they get their token and messages when 5 people, 2 people and 0 people are ahead of them (section 16.11). Off by default.

5 The staff workspace

Nurses, doctors and administrators use one workspace at `/staff`. It works on laptops, tablets and phones; on a phone the sidebar becomes a top bar.

Signing in

- Staff choose their **hospital** from a picker (the server lists hospital names only), then enter email and password. Passwords are at least 12 characters and stored as Argon2id hashes.
- Five wrong passwords lock the account for 15 minutes; sign-in is also rate-limited per address. Every success, failure and lockout is in the audit log.
- A successful sign-in gives a **10-minute access token** (kept in memory) and an **8-hour refresh cookie** (HttpOnly, SameSite=Strict, scoped to the sign-in path). The browser renews the access token silently, so a shift needs one sign-in.
- A deactivated or deleted account loses access within 10 minutes, and an open queue board closes within a minute.
- **Use another account** lets a second person sign in on a shared PC; each browser tab keeps its own account.

What each role sees

Page	Nurse	Doctor	Administrator
Queue (live dashboard, home page)	✓ own department	✓ own department	✓ every department
Consultations (case sheets)	read via queue/register	✓	✓
Patients (register, assisted registration)	✓	✓	✓
Reports	–	–	✓
Settings	–	–	✓

The header shows the signed-in person, their role and a **Live** badge for the live connection (it turns to **Reconnecting** when the connection drops, and reconnects by itself, renewing an expired sign-in first). A role that opens a page it may not use is sent back to the queue; the server refuses the data anyway.

6 Queue dashboard (nurses and doctors)

Today's waiting room for one department, in fair calling order, updating live. It is the home page for nurses and doctors.

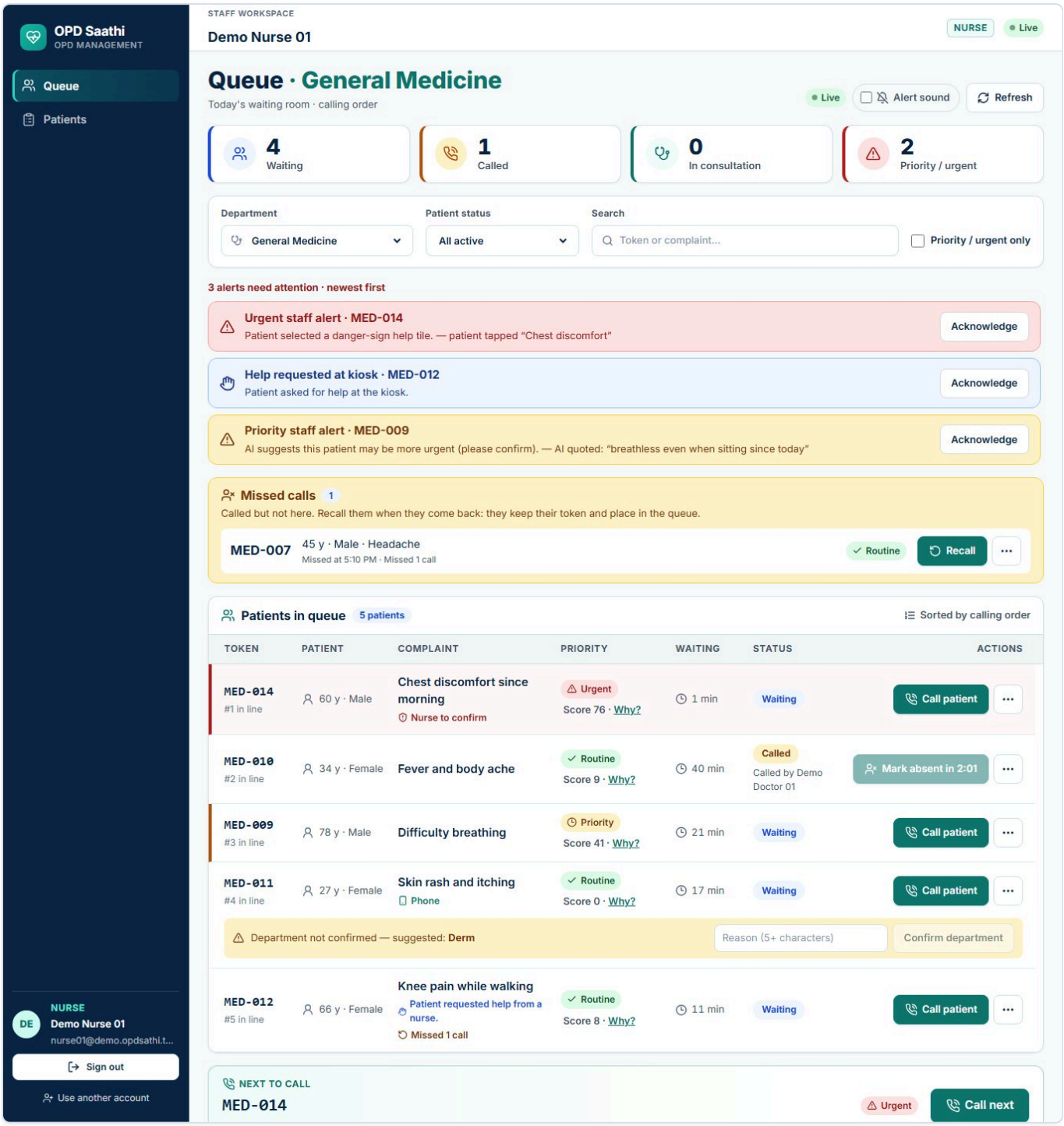


Figure 6.1 The queue board for General Medicine: counters, filters, three kinds of alert, the missed-calls panel and the patients in calling order (synthetic data).

What is on the page

Part	What it shows and does
Counters	Waiting, Called, In consultation, Priority/urgent. Clicking one filters the list.
Filters	Department (staff see their own; admins any), patient status, search by token or complaint, "Priority / urgent only".
Alerts	Newest first, each with an **Acknowledge** button and an optional alert sound: • Urgent a danger sign or an urgent red-flag rule, with what the patient tapped or said; • Help requested the patient pressed "Call staff" (a patient who stays 90 seconds on the review screen instead gets a "Patient requested help" badge on their row); • Priority an AI suggestion that the patient may be more urgent, always quoting the patient's words, for a nurse to confirm. Unacknowledged urgent alerts are re-raised every 60 seconds.
Missed calls	Patients who were called and did not come. They keep token and place; **Recall** puts them back (chapter 16.7).
Patients in queue	Token and place, age and sex, complaint in English, priority with the triage score and **Why?**, waiting time, status (with "Called by …"), and badges: Nurse to confirm for an unconfirmed danger sign, Phone for phone check-ins, "Patient requested help", "Missed 1 call".
Department not confirmed	When automatic routing was unsure, the row shows the suggestion; a nurse types a reason and confirms or changes the department.
Next to call	Who the server will call next, with **Call next**. Two doctors pressing it together get two different patients.

The triage score, explained on the page

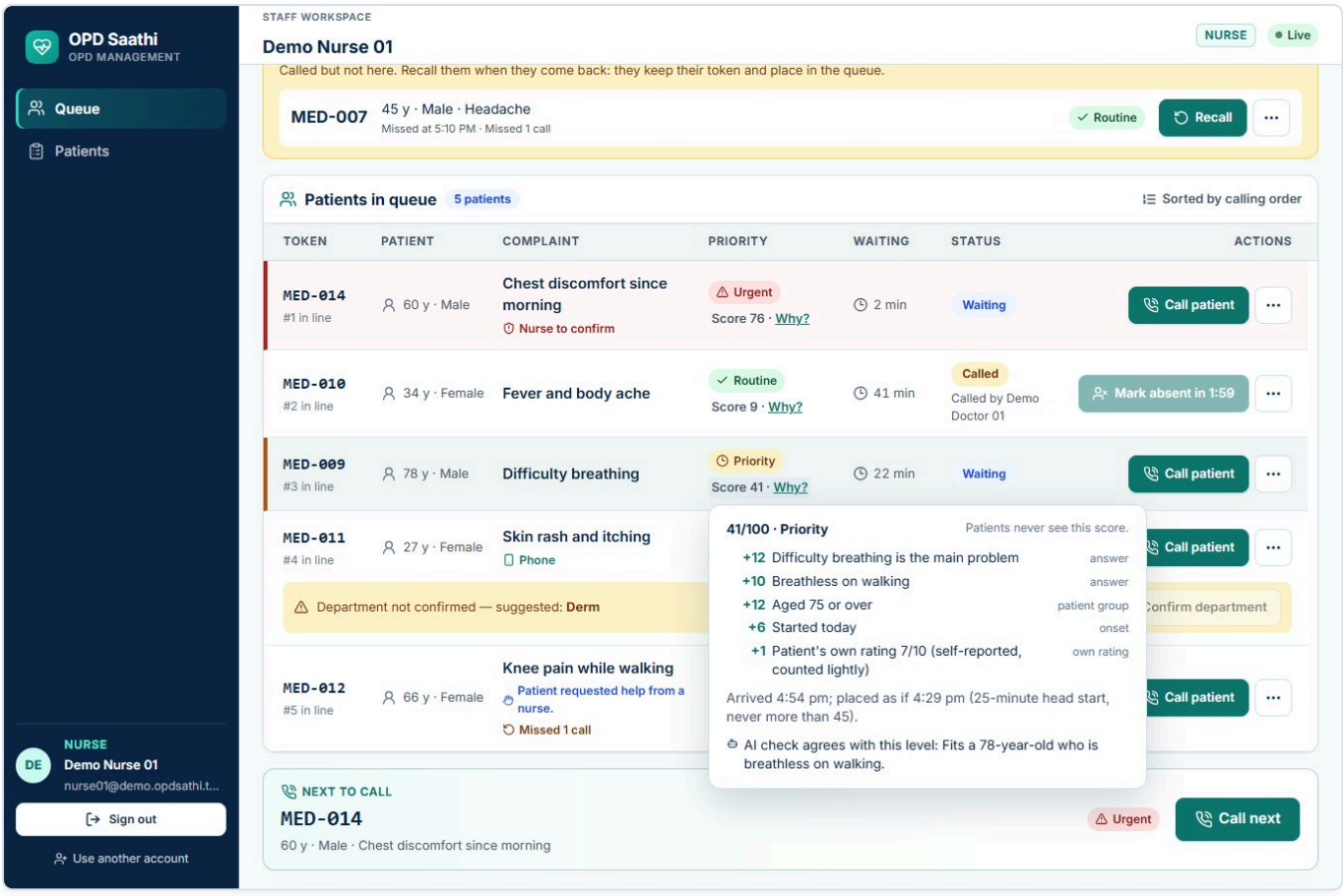


Figure 6.2 "Why?" lists every point of the score, the head start it gave, and the AI check. Patients never see this.

The score (0–100) adds points for danger signs and red flags, warning answers, age and pregnancy, recent onset, and, lightly, the patient's own 0–10 rating. Urgent patients go first; everyone else keeps arrival order with a head start of 0.6 minutes per point, never more than 45. The full rules and a worked example are in sections 16.3 and 16.4.

Actions

Action	What happens
Call patient / Call next	A doctor's call reserves the patient for that doctor (one patient per doctor at a time); a nurse's call does not reserve. The status page and the board update live.
Start / Complete	Doctors start and finish the consultation (also from the consultation page).
Mark absent	Available after the hospital's grace period (3 minutes by default; the button counts down). The patient keeps token, urgency and arrival time and moves to Missed calls.
Recall	Back into the queue on the original arrival time, so normally next within their priority class.
Put back (⋯)	Undo a call made by mistake, with no missed call on record.
No-show (⋯)	Only for a patient already marked absent who has left.
Change priority / department (⋯)	Nurses and admins, with a written reason; recorded in the audit log. Urgency never drops without such a decision.
Acknowledge	Clears an alert and, for a danger sign, removes "Nurse to confirm".
Open case sheet	Opens the consultation workspace for that visit.

Every action carries an `Idempotency-Key`, so a double click or a retried request never repeats it, and every action is checked on the server against the staff member's department.

Who can do what

Action	Doctor	Nurse	Administrator
Call a patient, call next	Yes (reserves the patient)	Yes (no reservation)	Yes
Start, complete a consultation	Yes, unless another doctor has the patient	No	Yes
Mark absent, put back	Yes, unless another doctor has the patient	Yes	Yes
Recall, mark no-show	Yes	Yes	Yes
Cancel a visit	No	Yes	Yes
Change priority or department (with a reason)	No	Yes	Yes
Link a returning patient at the desk	No	Yes	Yes
Set the grace period	No	No	Yes

Everything is limited to the staff member's own department; a visit in another department is reported as "not found". Administrators act across the hospital. Source: `docs/QUEUE_POLICY.md`.

Linking a returning patient at the desk

A patient who forgot their OPD card can be linked by a nurse: search by OPD number or phone number (results show OPD number, age, sex, visit count and last visit date, never names or clinical details), confirm identity in person, then link. A record created at check-in for someone who registered as new is emptied and its number retired. Linking is refused when both records already have history: there is no automatic merge. The search (`patient.lookup`) and the link (`queue.patient_linked`) are audited, and the doctor's view shows that staff made the link.

Live and reliable

- Updates arrive over a WebSocket within a second; the board renders a 5,000-patient list and applies an urgent change in well under 100 ms (virtualised rows).
- A board whose sign-in expires, or whose account is deactivated, is closed by the server (4401/4403) and reconnects after renewing the sign-in.
- Visits left open on an earlier day are closed automatically after midnight, so every morning starts with an empty queue (section 16.14).

7 The doctor's consultation workspace

Left: today's patients. Right: one patient's case sheet, organised into tabs so the doctor sees what matters first: red flags, the AI draft to check, the complaint, then the details one click away.

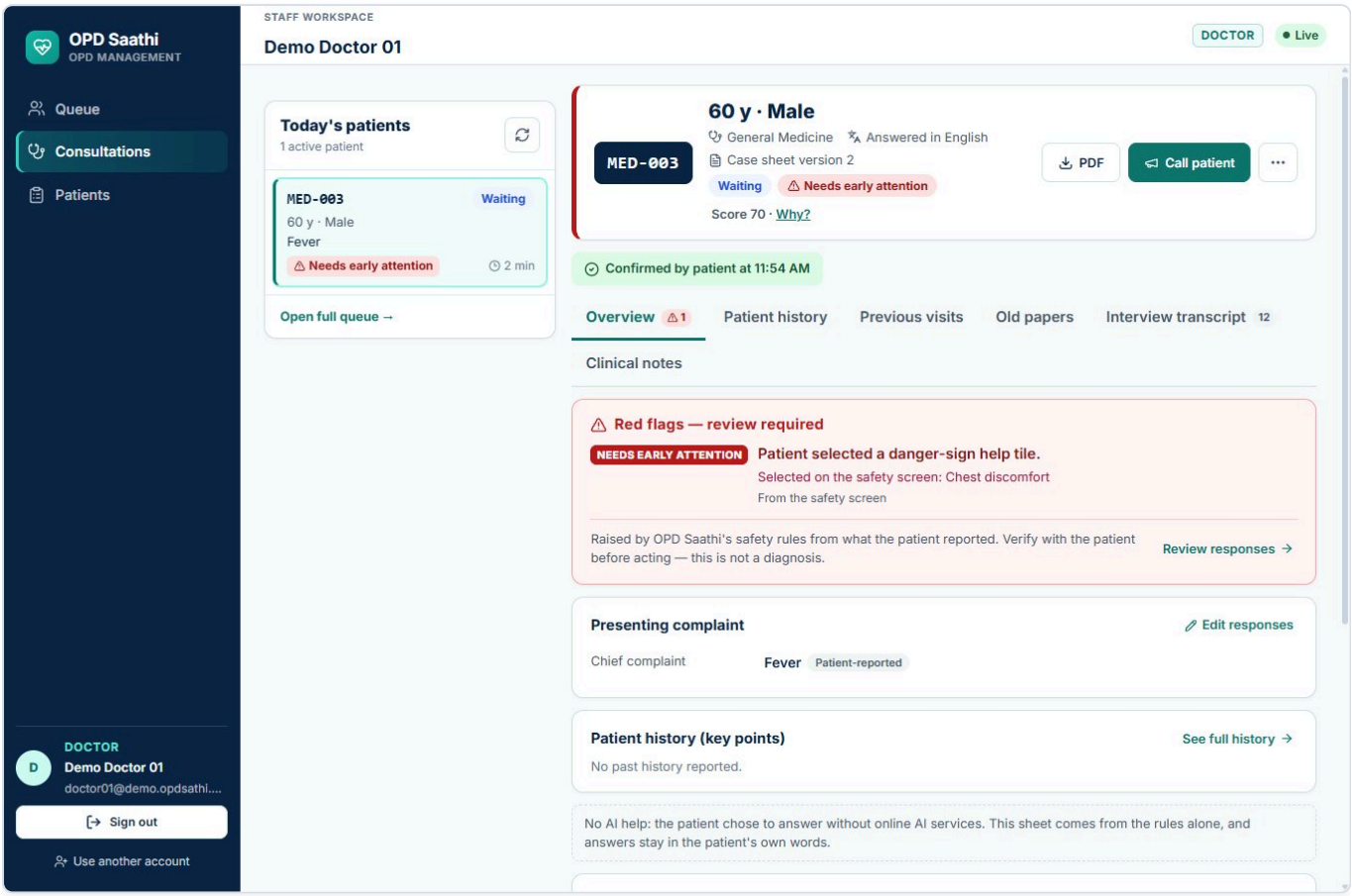


Figure 7.1 An urgent patient: the red-flag card says what was reported and where it came from, and that it is not a diagnosis. This patient said no to AI help, so the sheet comes from the rules alone.

7.1 The header (always visible)

- Token, age and sex, department, the language the patient answered in, the case-sheet version, status and urgency pills, the triage score with **Why?**.
- One main action that follows the visit's real state: **Call patient** → **Start consultation** → **Finish consultation**; **Mark absent in 3:00** counts down after a call. **PDF** and a ⋯ menu (print, keyboard shortcuts `N` back to the queue, `E` edit, `P` print; stop AI help at the patient's request).
- A green "Confirmed by patient at 11:46" and, separately, a violet notice when AI organised some fields. Patient confirmation never counts as staff checking the AI.

7.2 Overview and the AI draft

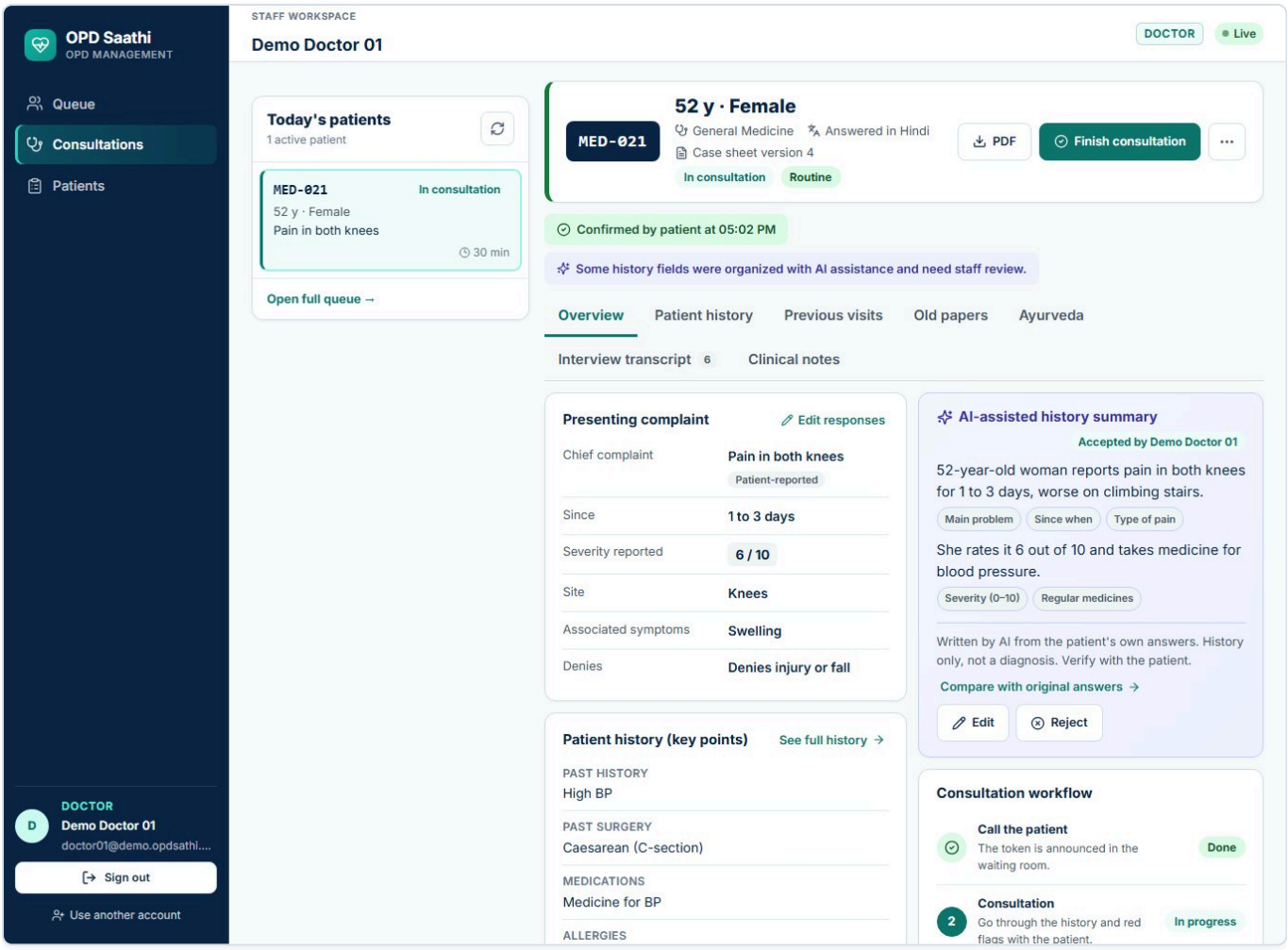


Figure 7.2 Each sentence of the AI summary carries chips for the answers it rests on (Main problem, Since when, Type of pain…). Here the doctor has accepted it.

How the AI draft is reviewed

- The model writes the brief as sentences, each citing the `question_id`s of the answers that support it. The server keeps only citations to questions this patient was actually asked and drops any sentence left without one; the output guard checks every sentence.
- A chip opens that answer in the transcript, highlighted. "Compare with original answers" does the same for the whole brief.
- The doctor records **Accept**, **Edit** (their own words; the AI wording is kept for the record) or **Reject**. The decision is versioned (`If-Match`), audited and limited to doctors of the treating department.
- The PDF says who reviewed it and never prints a rejected draft. A background AI step never overwrites a reviewed summary. If the patient withdraws AI consent, the draft disappears and cannot be reviewed.

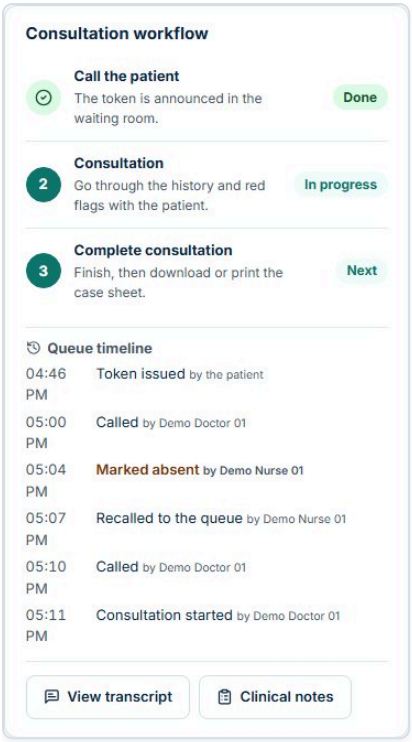


Figure 7.3 The workflow card and the queue timeline.

Below the draft come the **presenting complaint** (complaint, since when, severity, site, character, associated symptoms, denials) and the **key history points** (past illnesses, operations, medicines, allergies, family, tobacco and alcohol, open questions), each with "See full history".

The **consultation workflow** card follows the visit's real state: call the patient → consultation → complete. Under it, the **queue timeline** lists every change to the visit, who made it and when: token issued by the patient, called by the doctor, marked absent by a nurse, recalled, called again, consultation started. The timeline is append-only: the application can add entries but never change or delete them (`GET /visits/{id}/queue-events`, staff of the visit's department only).

When the patient said no to AI help, the overview says so plainly: the sheet comes from the rules alone and the answers stay in the patient's own words (figure 7.1).

7.3 Patient history: every field, with its source

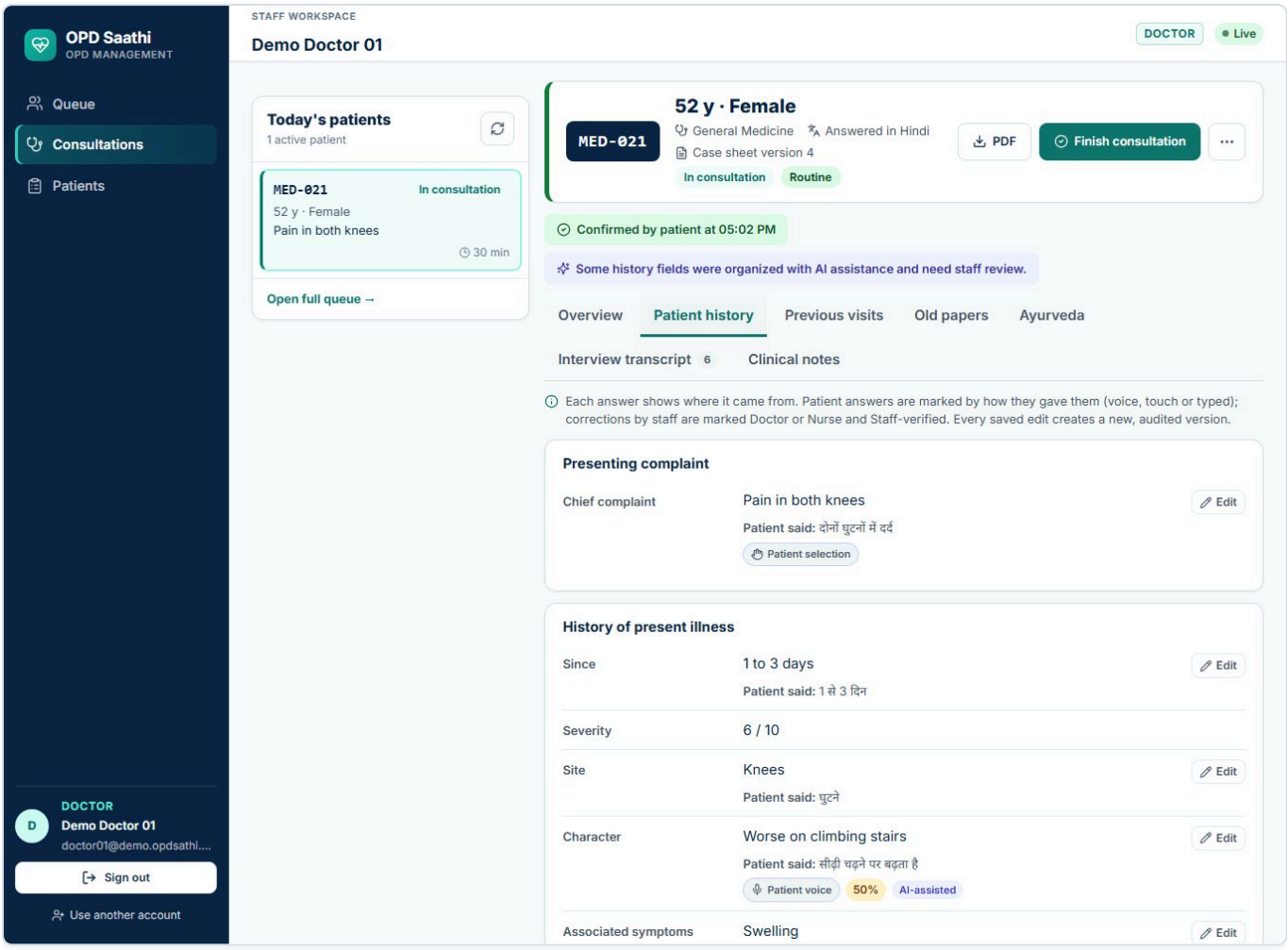


Figure 7.4 Each field shows the English text, what the patient said in their language, and how they gave it (selection, voice with recognition confidence, typed), with an AI-assisted tag where relevant.

- Sections: presenting complaint; history of present illness (since, severity, site, character, associated symptoms, denials); past illnesses and operations; medicines; allergies; family history; tobacco, alcohol and other habits; pregnancy where asked; open questions ("Does not know: …", "Preferred not to say: …").
- **Edit** any field. A save sends a restricted JSON patch with the version it was based on; every save creates a new version and an audit entry, and staff edits are marked Staff-verified. If two people edit at once, the second is told and can reload or re-apply their change; nothing is silently overwritten. Leaving with an unsaved edit asks first.

7.4 Previous visits

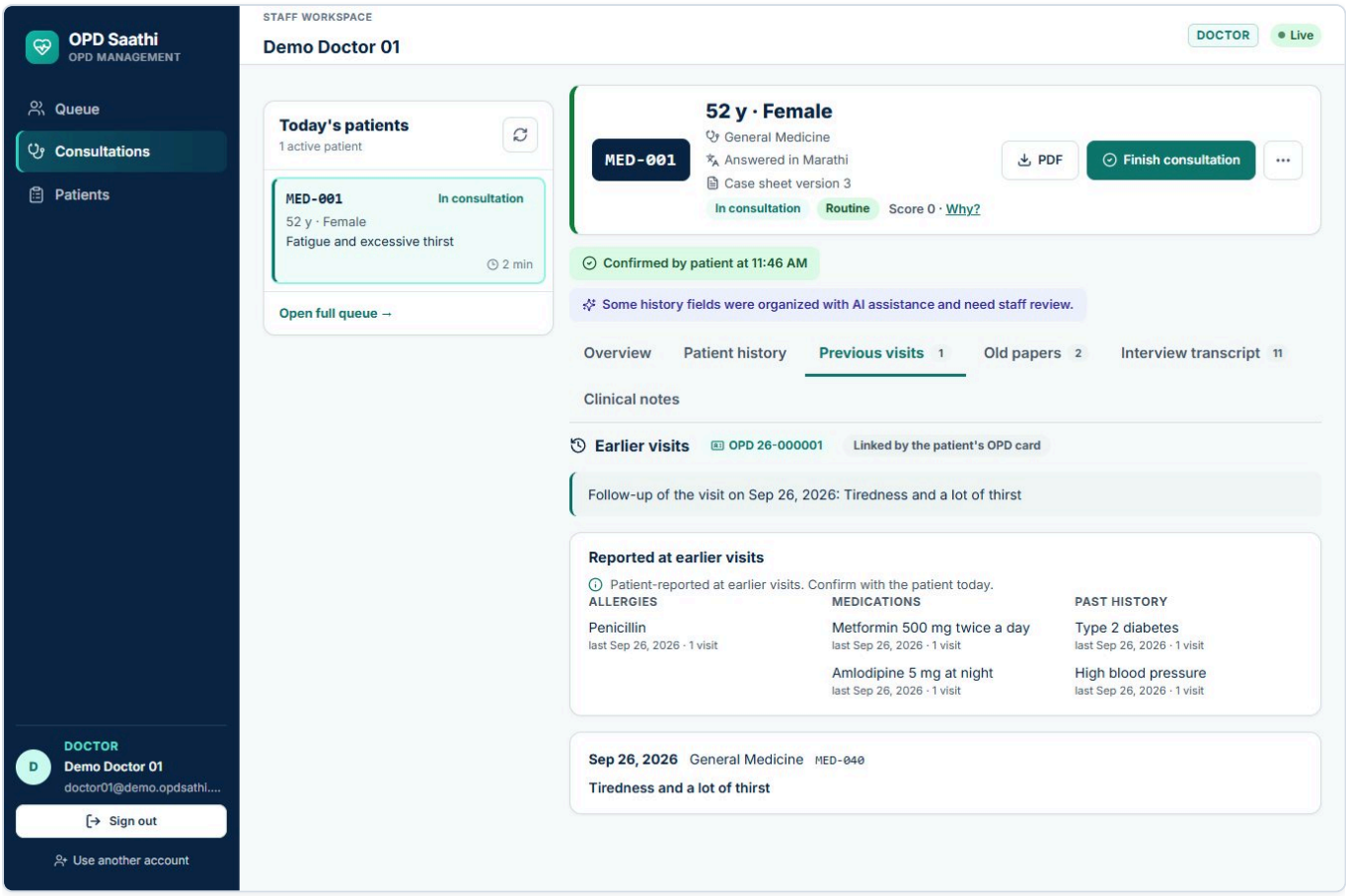


Figure 7.5 A returning patient linked by her OPD card: today's visit follows the one on 26 September; what she reported earlier is listed with the date it was last reported.

Up to 20 earlier visits at the same hospital: date, department, token, complaint, red flags and the AI brief (marked when staff edited the sheet), plus allergies, medicines and conditions with the date each was last reported, labelled "patient-reported, confirm today". Doctors and nurses see it only through a visit of their own department that is still open or took place today; once that visit is closed and in the past the server answers `HISTORY_CLOSED`. Administrators can read it through any visit. Every read is audited (`patient_history.read`). Records are never shared between hospitals.

7.5 Old papers

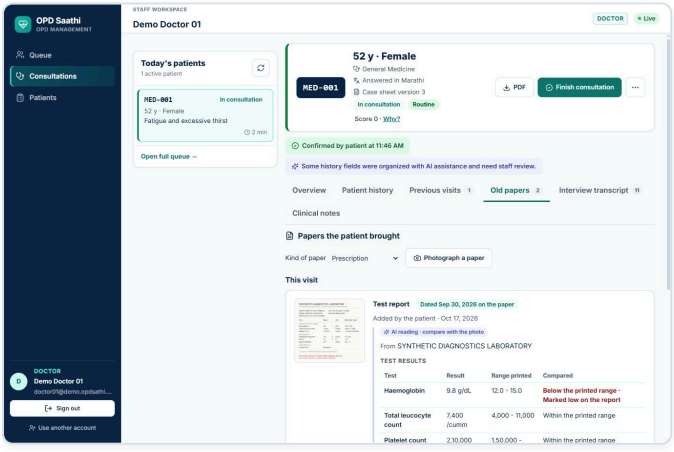


Figure 7.6 The test report the patient photographed, beside what AI read from it.

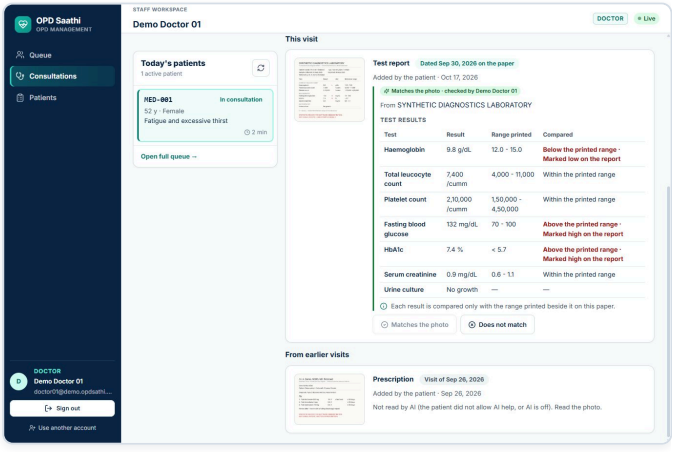


Figure 7.7 Accepted by the doctor; results compared with the printed ranges; an earlier paper kept unread.

- Papers from this visit and, while the patient is under care, from earlier visits, newest first by the date printed on them. A doctor or nurse can photograph more (**Kind of paper → Photograph a paper**), up to 10 per visit.
- With AI consent, each paper has an **AI reading**: the source (laboratory or doctor), diagnoses and medicines as written, test results with value, unit and the range printed beside them, procedures and the printed date.
- **Compared** is plain arithmetic against the printed range on the same paper: "Below the printed range", "Within", "Above", plus a high/low mark printed by the laboratory. Nothing is flagged from a general range or by the model (section 16.10).
- The doctor records **Matches the photo** or **Does not match** (doctors only, changeable, audited). A rejected reading is no longer shown. Without AI consent the photo is kept unread: "Not read by AI. Read the photo."
- Photos are sent with `Cache-Control: no-store`, and every view of a photo is audited.

7.6 Ayurveda

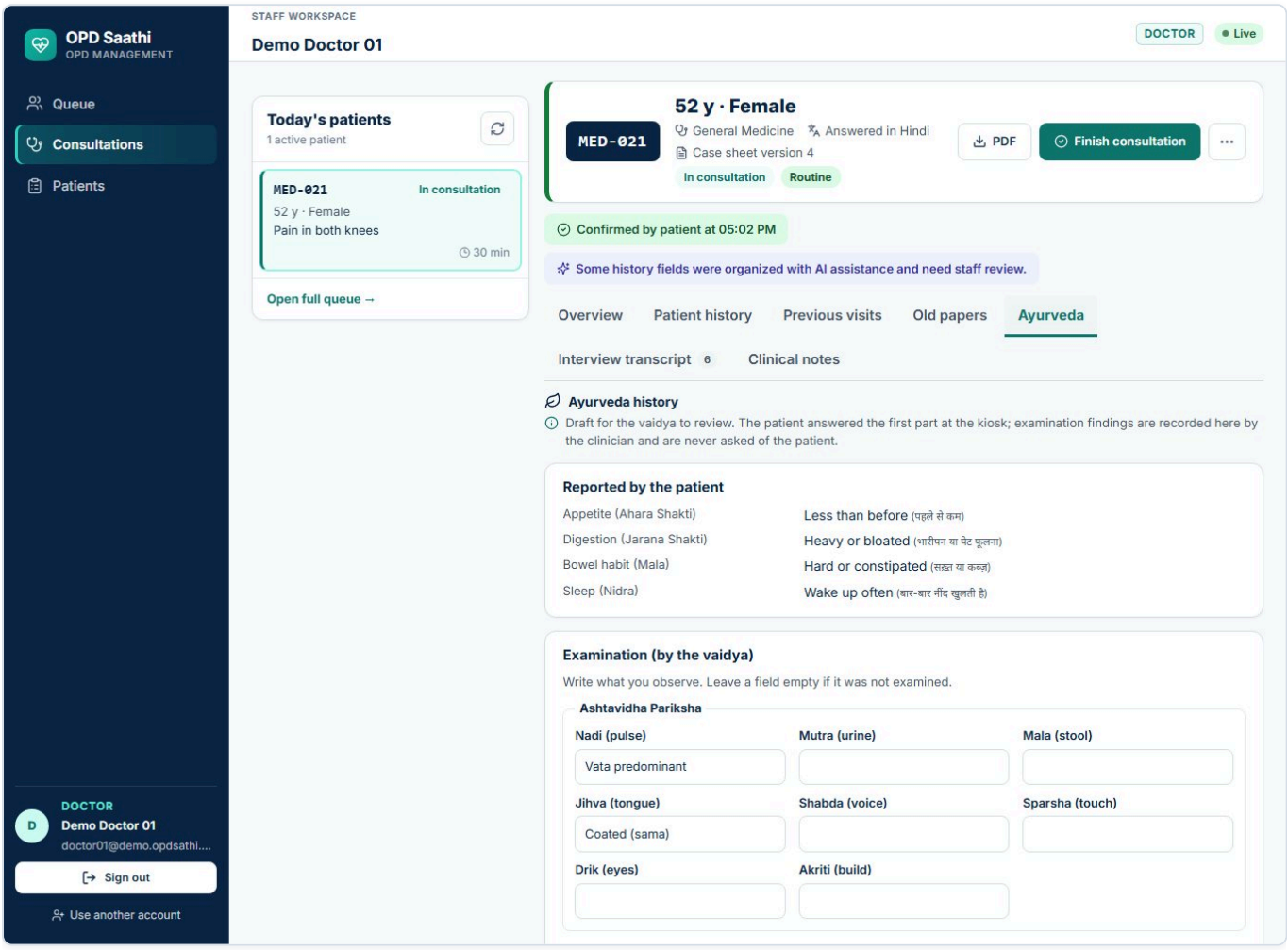


Figure 7.8 When the hospital turns the Ayurveda module on: what the patient reported, and the examination fields for the vaidya.

The patient answers six questions at the kiosk: appetite (Ahara Shakti), digestion (Jarana Shakti), bowel habit (Mala), sleep (Nidra), capacity for exertion (Vyayama Shakti) and diet. The clinician records 16 examination findings that are never asked of the patient: Ashtavidha Pariksha (Nadi, Mutra, Mala, Jihva, Shabda, Sparsha, Drik, Akriti) and Dashavidha items (Prakriti, Vikriti, Sara, Samhanana, Pramana, Satmya, Sattva, Vaya). Everything is labelled a draft for the vaidya's review.

7.7 Interview transcript and clinical notes

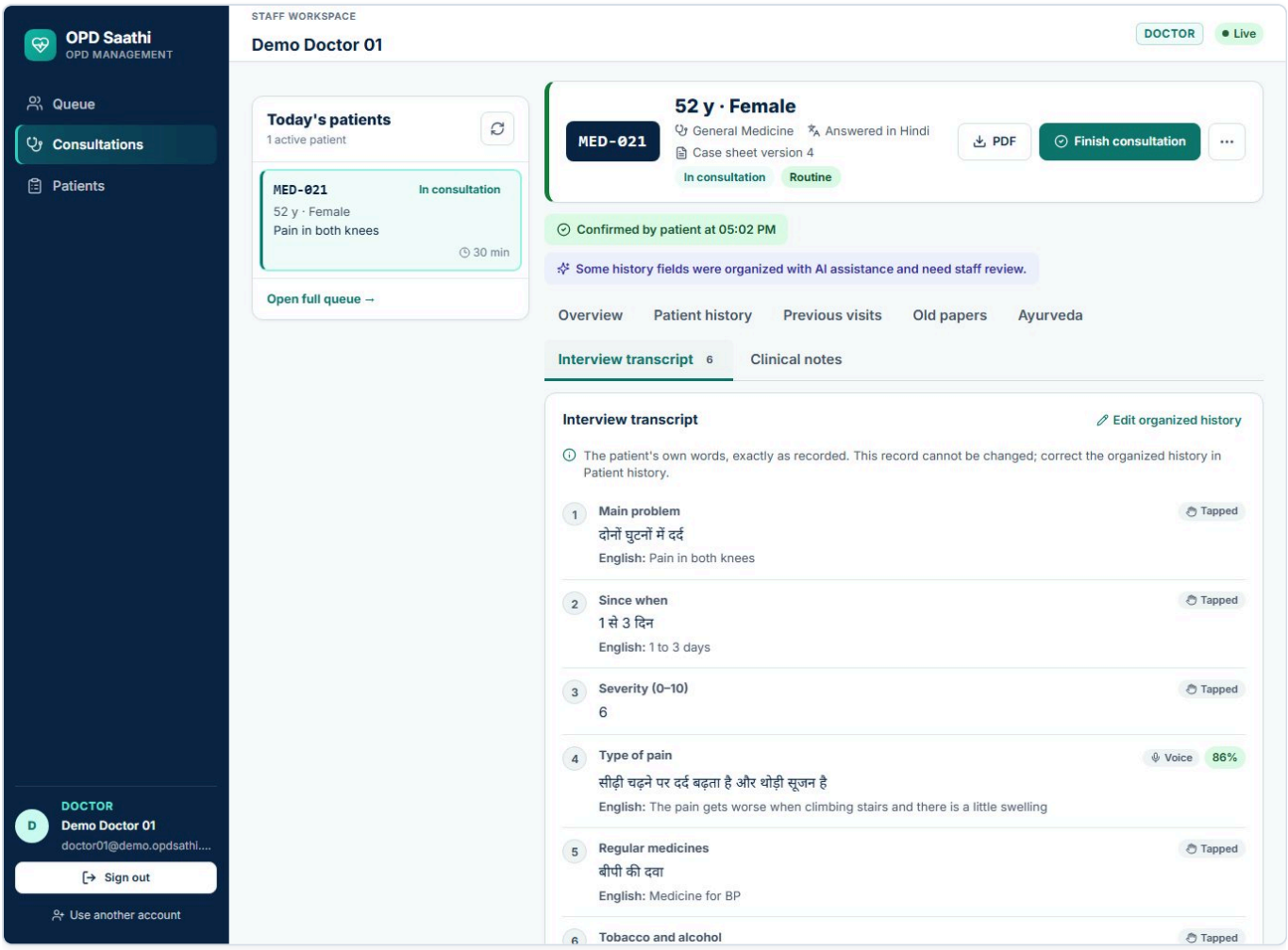


Figure 7.9 The transcript: every question and the patient's exact answer in their language, with English, how it was given (tapped, voice with confidence, typed) and "AI follow-up" labels. This record cannot be changed.

Clinical notes lists staff edits to this case sheet, the version and its audit note, and the corrections the patient made before confirming. Examination findings, assessment and plan are not stored in OPD Saathi yet; doctors record them in the hospital's own record (Phase 7 task 7.11.1).

7.8 The PDF case sheet

PDF or **Print** always produces the full case sheet, whichever tab is open: a compact, multilingual, history-only document rendered on the server (WeasyPrint, with Inter and Noto fonts so Hindi and Marathi print correctly). It includes the reviewed AI summary with the reviewer's name (never a rejected draft), open questions, the Ayurveda section when present, and the disclaimer "History only. Not a diagnosis."

8 Patients register and assisted registration

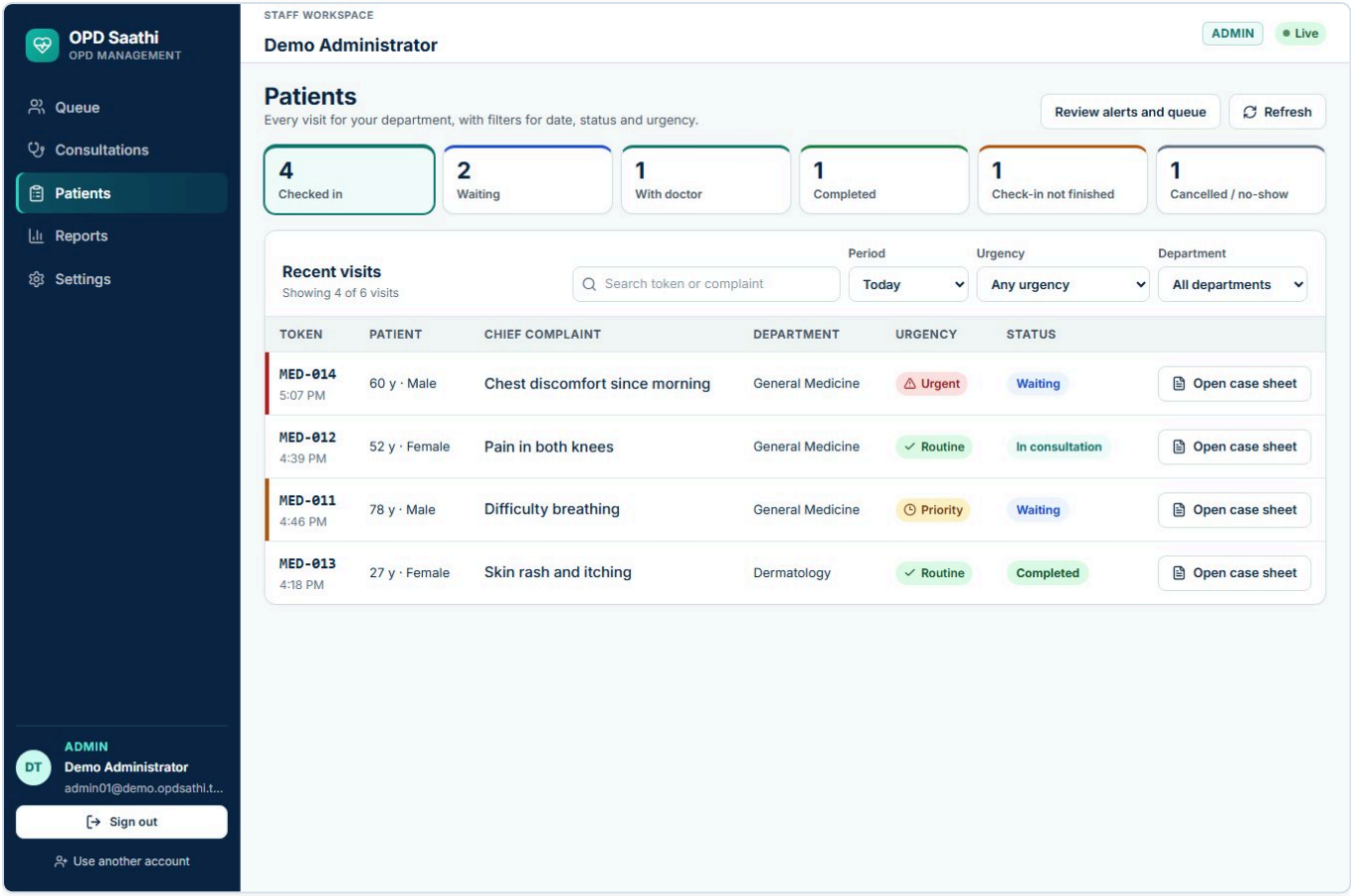


Figure 8.1 Every visit in a period, not just today's waiting room, with "Open case sheet" on each row.

- **Status tiles:** Checked in, Waiting, With doctor, Completed, Check-in not finished, Cancelled / no-show. Clicking filters the list.
- **Filters:** period (today, 7 days, 30 days), urgency, department, search by token or complaint.
- **Assisted registration:** a nurse starts a check-in for a patient who cannot use the kiosk (elderly, unwell, unable to read). Consent is always shown first and recorded with the channel `nurse`; staff cannot skip it. The visit then follows the same interview, safety rules and queue as a kiosk check-in.

9 Reports

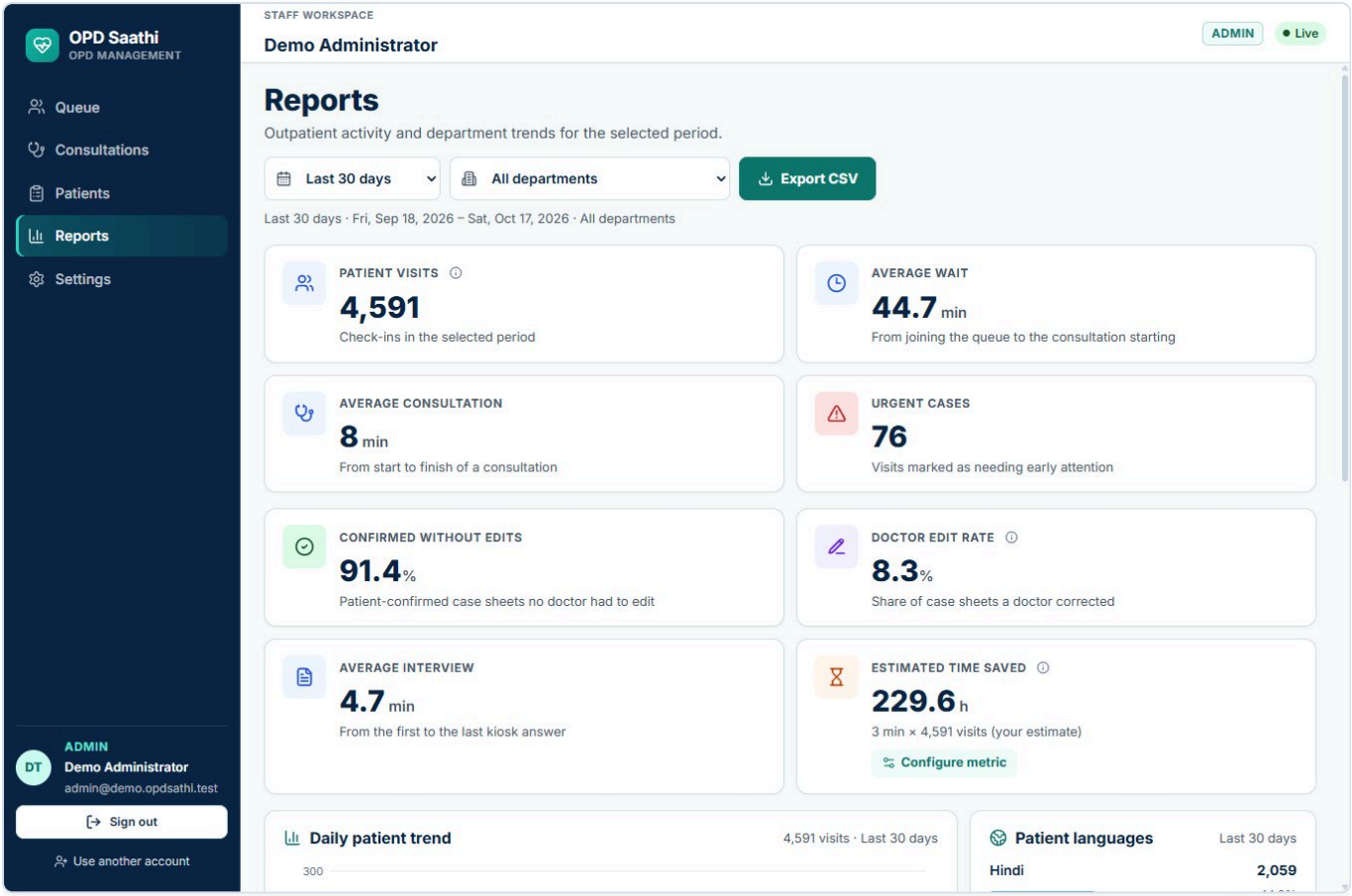


Figure 9.1 Reports for the last 30 days of the demo hospital (a month of synthetic history).

Filters (Today, 7, 30 or 90 days, and department) control every card, the chart and the CSV export, so all numbers describe the same period.

Card	Meaning
Patient visits	Every check-in in the period, including patients not yet seen. A patient who comes twice counts twice.
Average wait	From joining the queue to the consultation starting.
Average consultation	From the start to the finish of a consultation.
Urgent cases	Visits marked as needing early attention.
Confirmed without edits	Of the case sheets patients confirmed, the share no doctor needed to change: a measure of how accurate the kiosk history is.
Doctor edit rate	Of visits that produced a case sheet, the share a doctor corrected.
Average interview	From the first to the last kiosk answer.
Estimated time saved	Visits × the minutes of history-taking each interview is assumed to save (an administrator sets it, 0–60 minutes, default 3; every change is audited). Clearly labelled an estimate.

- **Daily patient trend** (one bar per day, zero days shown, exact values on hover, tap or arrow keys; its total always equals the visits card), **patient languages**, and **department load** (visits, urgent cases, average wait, share).

- **Export CSV** of the same period and department. Cells that begin with `=`, `+`, `-`, `@`, a tab or a carriage return are prefixed with an apostrophe so a spreadsheet cannot run them as formulas.
- "No data" (nothing to measure) is shown differently from a real zero. Figures come from a daily rollup refreshed every 5 minutes (section 16.12); every view of the reports is audited.

10 Settings and administration

Administrators run the hospital from Settings, with three tabs: Users, Kiosks and Departments. No commands are needed day to day.

10.1 Users

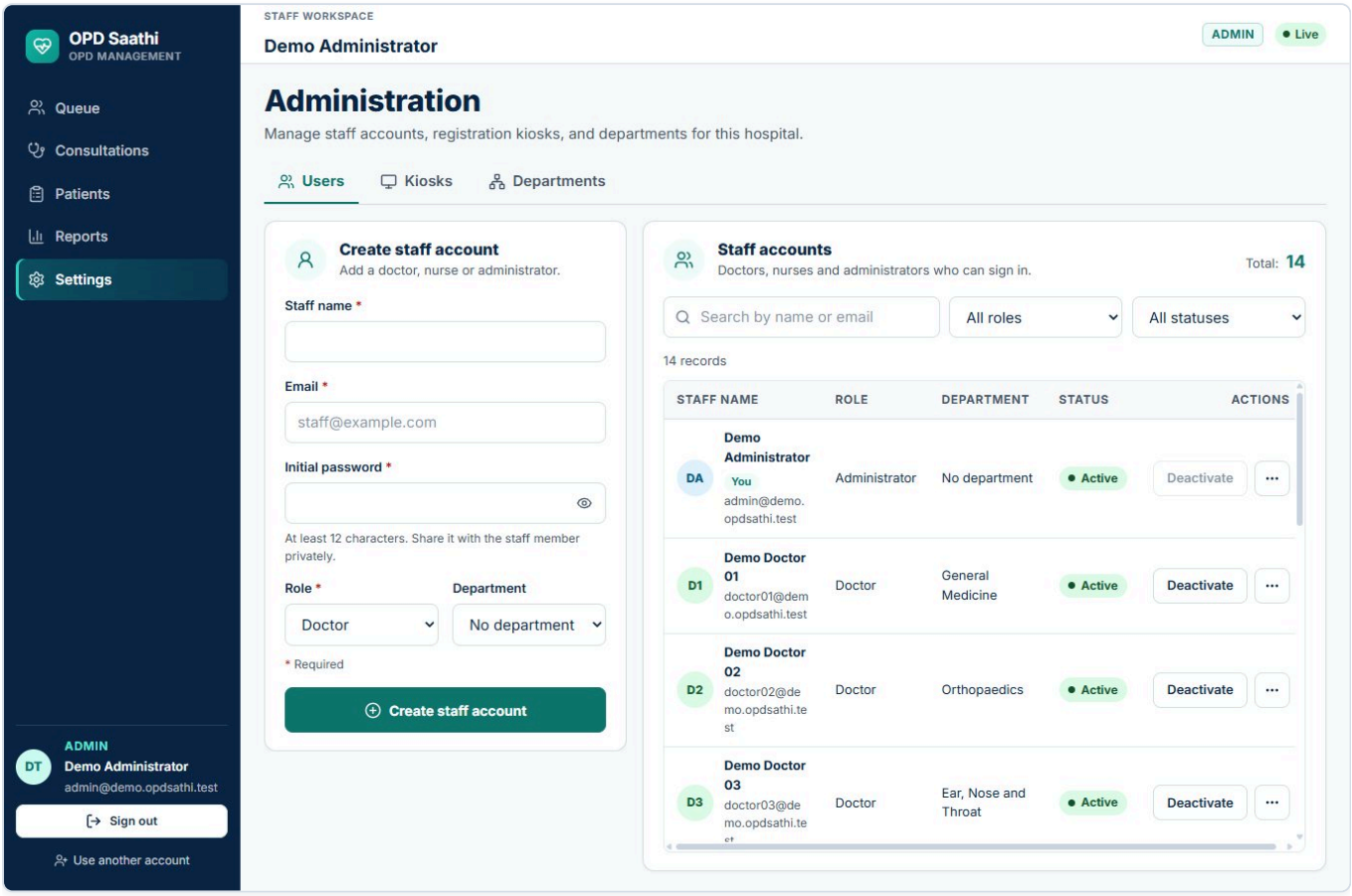


Figure 10.1 Create an account on the left; search, filter and manage staff on the right.

- Create doctors, nurses and administrators: name, email, an initial password of 12+ characters (given to the person privately), role and department.
- **Deactivate** asks first; the person is signed out within 10 minutes and can be activated again. **Delete** (⋯) removes the account for good: their past work stays in patient records and the email can be reused. Nobody can deactivate or delete themselves.
- Not yet in the screen: editing a person's name, email or department (the API supports it), resetting a password, and the hospital AI switch (API only). These are Phase 7 tasks 7.3.1, 7.3.3 and 7.11.5. Until then: delete and re-create the account; `create-admin` recovers a locked-out administrator.

10.2 Kiosks and phone check-in QRs

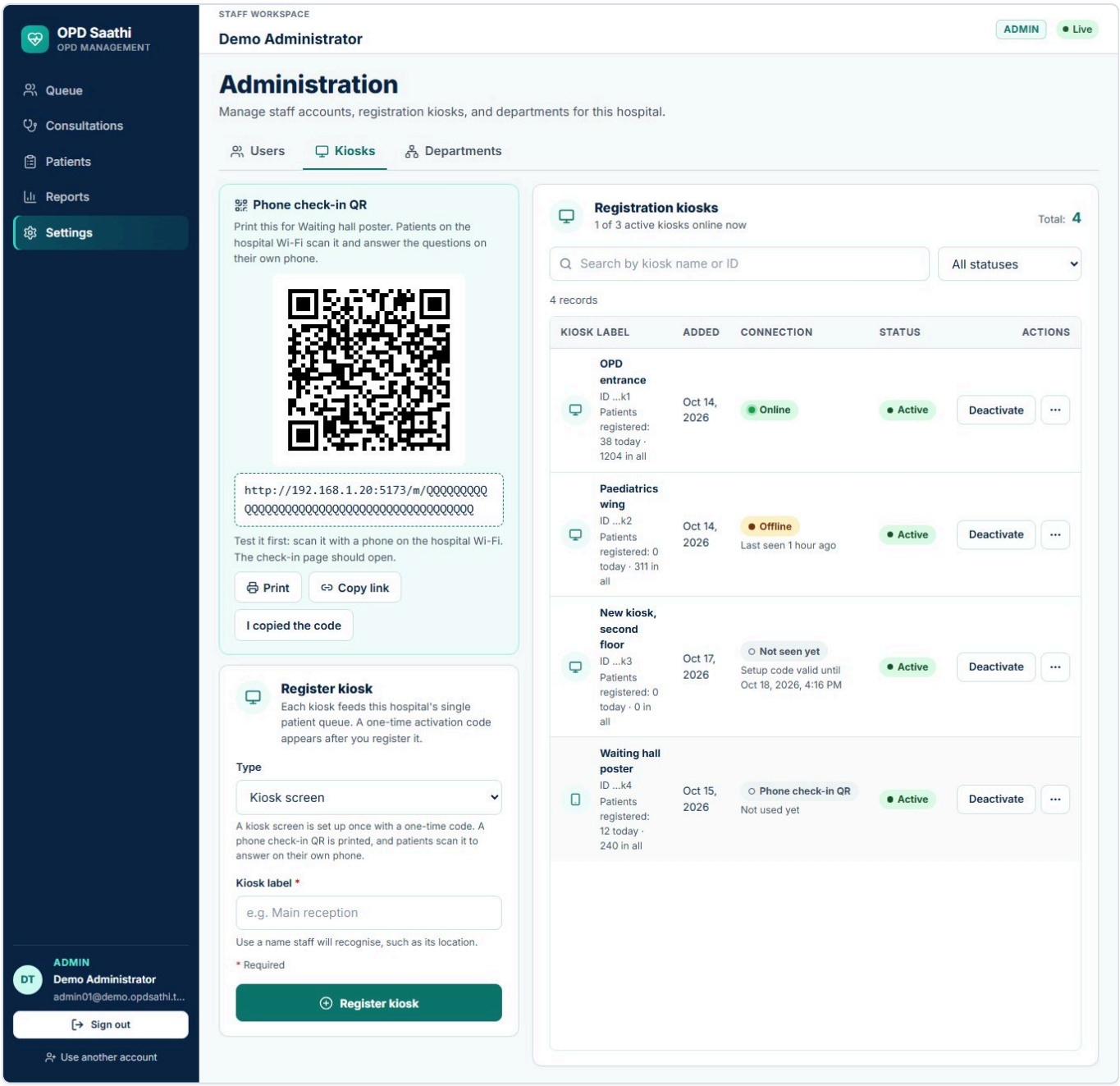


Figure 10.2 Kiosks: an online kiosk, an offline one, one waiting for its setup code, and a phone check-in QR shown for printing.

- **Type**: *Kiosk screen* (paired once with a one-time code) or *Phone check-in QR* (printed; patients scan it).
- For each kiosk: connection (Online, Offline with "last seen", Not seen yet with "setup code valid until…", Turned off), "Patients registered: N today · M in all", and the ⋯ menu.
- **New activation code**: a fresh 24-hour code for the same kiosk record; the current device keeps working until the new code is used. **Show QR** and **Replace QR** for phone points; **Print** and **Copy link**.

Action	What happens	To undo
Deactivate	The kiosk shows "This kiosk is turned off" and cannot start check-ins; it stays paired.	Activate: works at once.
New activation code	A new device can take over; the old one is signed out once it does; history kept.	Not needed.
Replace QR (phone)	Old posters stop working at once; print the new QR.	Not possible; print the new one.

Action	What happens	To undo
Delete	The device stops working immediately and leaves the list; patients it registered stay in their records.	Register a new kiosk.

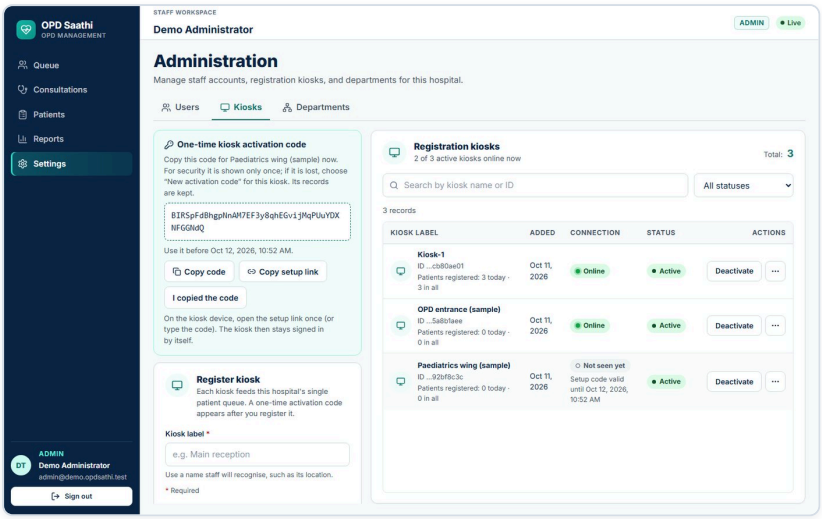


Figure 10.3 A one-time activation code with **Copy setup link** and the time it must be used by.

10.3 Departments, calling patients and the patient interview

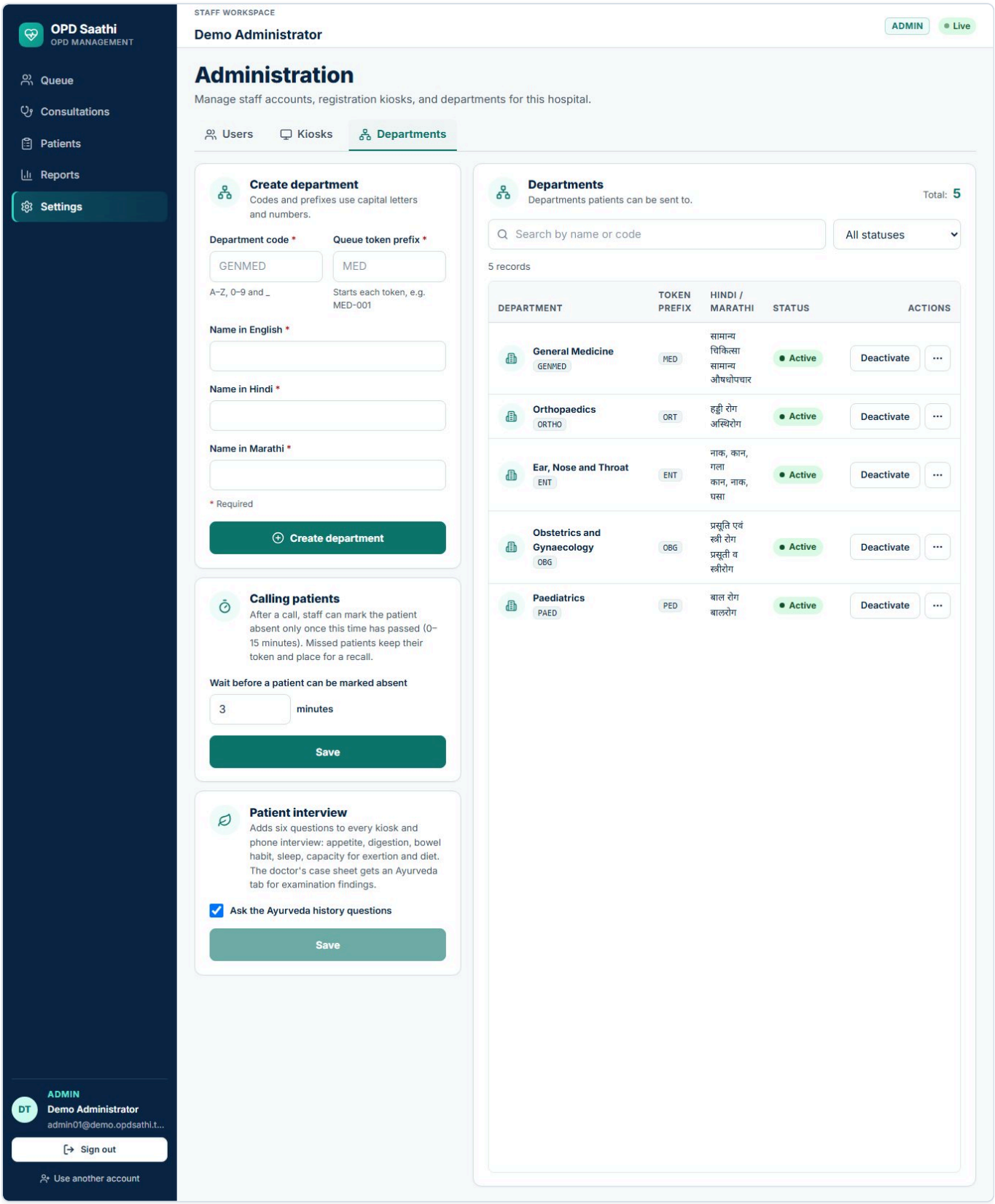


Figure 10.4 Departments with token prefixes and Hindi/Marathi names; "Calling patients" sets the grace period; "Patient interview" switches the Ayurveda history on.

- A department has a **code** (capitals, digits and `_`, e.g. `CARDIO`), a **token prefix** of up to 8 capitals or digits (`CAR` gives `CAR-001`), and names in English, Hindi and Marathi.

- Automatic routing understands the standard codes `GENMED ORTHO ENT OBG PAED DERM OPHTH SURG`; other departments receive patients sent by staff.
- **Deactivate** stops new patients at once. **Delete** is allowed only when no visit is open there and no staff member is assigned; the message says what to do instead.
- **Calling patients:** how long after a call staff must wait before marking a patient absent (0–15 minutes, default 3). **Patient interview:** adds the six Ayurveda questions to every kiosk and phone interview and an Ayurveda tab to the case sheet; it applies to interviews that start afterwards. Both changes are audited.

> **Delete never erases history**
>
> Visits, case sheets and the audit log keep pointing at a deleted staff member, kiosk or department, so records stay complete. The record leaves Settings and every choice list, and its email, code or prefix can be reused.

11 AI in OPD Saathi

AI makes OPD Saathi friendlier and faster, but it is optional twice over: the hospital must switch it on, and each patient must say yes. Without it, every core feature still works.

> **The AI gate**
> Language-model features need **both** the hospital's AI switch and the patient's consent to online AI help. Online speech recognition and speaking the patient's own words need the patient's consent. The administrator's switch can turn AI off for everyone; it can never turn it on for a patient who said no. The gate is enforced by the API on every call (`services/api/app/api/v1/ai_gate.py`).

Feature	With AI (patient said yes)	Without AI
Speech to text	Sarvam (best for Hindi/Marathi), else Gemini; the words appear for checking	Tap and type (and, only with consent, the browser's own recogniser)
Reading aloud	Natural Indian voices (Sarvam Bulbul or Gemini TTS) for the patient's own read-back	Recorded question audio (hospital text) and the device's voice
Follow-up questions	Up to 3 complaint-specific questions in three languages, validated, labelled	The fixed, logical bank flow
English case sheet	Every answer translated; a cited brief for the doctor to review	Tapped answers translated from the bilingual bank; spoken/typed words kept in the patient's language
Safety second opinion	The model may add a red flag, quoting the patient; it counts as priority at most until a nurse confirms	Rules engine only
Triage double-check	"AI agrees" or "AI suggests more urgent (please confirm)" next to the score	Score from the rules
Department suggestion	A configured-label classifier; disagreement with the rules goes to a nurse	Keyword routing
Old papers and prescriptions	Read into drafts the doctor accepts or rejects; prescription names confirmed by the patient	Photos kept unread for the doctor; prescription scan hidden
Routing, queue, tokens, reports	Always rules-based and identical with or without AI	

Guard-rails on every AI answer

- Requests ask for JSON that matches a schema; the answer is re-validated against the schema and the application's own models.
- An **output guard** rejects text that diagnoses ("you have…", "likely suffering from"), advises ("you should", "please take"), names medicines or doses, or contains links, in English, Hindi, Marathi and Hinglish. Rejected output is discarded and the deterministic result is used.
- Phone numbers, 12- and 14-digit identifiers (Aadhaar-like, ABHA) and email addresses are redacted before any request; names are never sent.
- At most 30 model calls per session and 1,024 output tokens per call; short timeouts; a circuit breaker after 5 failures; a fallback model; the patient never sees a provider failure (section 16.9).
- Prompts and responses are never logged; only counts and latencies are.

Providers today: Google Gemini (language model, also speech and reading photos) and Sarvam AI (Indian-language speech and voices). `PROVIDER_MODE=demo-safe` replays canned, non-clinical fixtures for offline rehearsals. Before real patient data, the hospital needs paid or enterprise provider agreements with data-processing terms (Phase 7 task 7.5.9).

12 One system, many hospitals

Recommended: one installation per hospital

- A server or PC inside the hospital runs PostgreSQL, Redis, the API, the worker and the web app with Docker.
- Patient data stays in the building; check-in and the queue keep working when the internet is down.
- Each installation has its own passwords and encryption keys.
- Installed with one script, `setup-hospital.ps1` (chapter 23).

Also possible: several hospitals on one server

- For example a district hosting its primary health centres: `setup-hospital.ps1 -AddHospital`.
- Staff pick their hospital at sign-in and see only their own.
- Separation is enforced by PostgreSQL row-level security, not only by the application: every tenant table refuses rows of another hospital.

How many kiosks?

As many as the hospital needs. Every kiosk and phone QR belongs to one hospital and feeds that hospital's department queues: 100 kiosks still mean one General Medicine queue, in fair calling order. Tokens are numbered per department per day (`MED-001` , `MED-002` …) whichever kiosk the patient used.

A second hospital was set up and checked live: its patient appeared only in its own queue, and neither hospital could open, call, change or delete the other's patients, staff or kiosks (20 of 20 checks). The automated isolation tests probe every route that names a record as another hospital, another department and another patient (section 19.5).

PART II

How it works inside

The architecture, the technology, every algorithm with its exact numbers, the data model, the API, security, offline behaviour and how quality is measured.

13. Architecture
14. Technology stack
15. Repository map
16. Algorithms and rules
17. Data model
18. API
19. Security and privacy
20. Offline use and the PWA
21. Quality: tests, CI and evaluations

13 Architecture

A local-first system: one React web app for every screen, a stateless FastAPI server, PostgreSQL as the source of truth, Redis for live state, and a background worker, all started with Docker Compose on one hospital server.

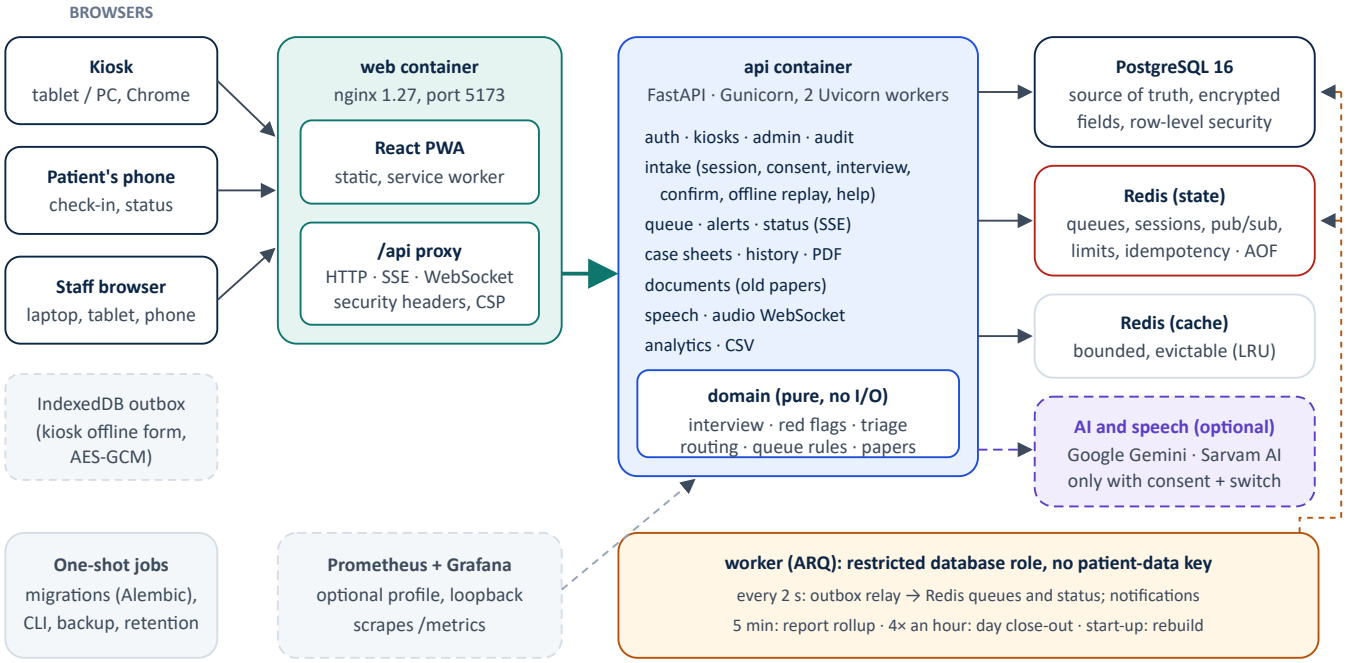


Figure 13.1 Runtime components of one OPD Saathi installation.

13.1 Components

Component	Responsibility
Web app (`apps/web`)	One React + TypeScript single-page app for the kiosk, the phone pages and the staff workspace, built with Vite, served as static files by nginx, installable as a PWA. Staff screens and their strings are lazy-loaded chunks, so the kiosk loads fast.
nginx (`infra/web`)	Serves the app with security headers and a strict CSP, and proxies `/api` to the API (HTTP, server-sent events and WebSockets), so kiosks, staff PCs and phones all need only one address and port.
API (`services/api/app`)	FastAPI with Pydantic models. Thin routers in `api/v1` ; all SQL in `db/repositories.py` ; pure, I/O-free rules in `domain` ; provider adapters and AI guard-rails in `ai` ; security primitives in `core` . Stateless: any number of processes can run.
PostgreSQL 16	Source of truth for every record: hospitals, staff, kiosks, patients, visits, consents, interview turns, case sheets, red flags, documents, the queue timeline, notifications, the outbox and the audit log. Row-level security per hospital; four database roles with least privilege.
Redis (state)	The queue read model (a sorted set per department), sequenced live deltas over pub/sub, intake sessions (2 h), rate-limit windows, idempotency records (24 h), per-session locks and the AI call budget. Durable (AOF every second, no eviction); the queue can always be rebuilt from PostgreSQL.
Redis (cache)	A separate, bounded, evictable cache (256 MB, LRU) for fixed generic text. Losing it loses nothing clinical.

Component	Responsibility
Worker	ARQ cron jobs: the outbox relay and notification delivery every 2 seconds, the analytics rollup every 5 minutes, the end-of-day close-out four times an hour, and a queue rebuild at start-up. It uses a restricted database role.

13.2 What happens when a patient confirms

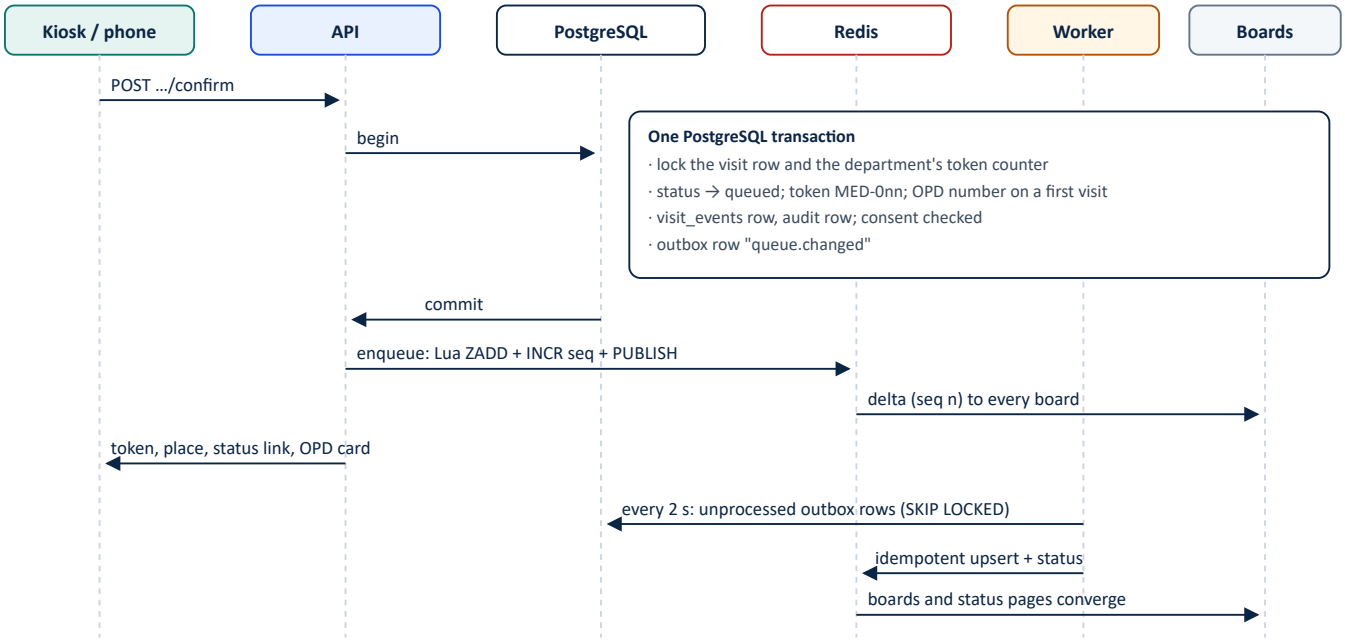


Figure 13.2 Confirmation issues the token in PostgreSQL first; Redis and the live screens follow. The worker's outbox relay repairs anything a crash in between could miss.

- **PostgreSQL first.** A token exists only once the transaction commits. A repeated or simultaneous confirmation returns the same token (the receipt is rebuilt from the database if the Redis session was lost).
- **The outbox pattern.** Every queue change writes an `outbox` row in the same transaction. The worker relays unprocessed rows every 2 seconds (`FOR UPDATE SKIP LOCKED` , up to 100 at a time), so the Redis read model converges even if the API crashed after committing.
- **Sequenced deltas.** Each department's queue has a sequence number incremented inside the same Lua script as the change; a board that sees a gap asks for a fresh snapshot.
- **Rebuildable.** If Redis is lost, the API or worker rebuilds every department's sorted set from today's queued visits at start-up (guarded by a lock), and `cli queue-rebuild` does it on demand.

13.3 Live updates

Channel	Used by	How
WebSocket `/ws/staff/queue`	Queue boards	Authenticates with the first frame (the staff token); then forwards queue deltas and hospital alerts for the staff member's department; re-checks the account every minute; re-raises unacknowledged urgent alerts every 60 s; closes 4401 at sign-in expiry and 4403 on deactivation.
Server-sent events `/status/{token}/stream`	Patient status page	Pushes the visit's state when its Redis status channel fires; the page falls back to polling `/status/{token}` (cached 2 s).
WebSocket `/ws/intake/{visit}/audio`	Kiosk voice answers	Streams microphone audio for live transcription (Sarvam realtime) with the patient's session; audio is never stored. Short clips can also be sent to `/transcribe` .

13.4 Design choices (architecture decision records)

ADR	Decision	Why
001	Locked stack: React/Vite/TS, FastAPI/Pydantic/SQLAlchemy, PostgreSQL, Redis, Docker Compose	One documented, testable path; no new framework, database or major library without an ADR and the owner's approval.
002	MVP substitutions behind production module boundaries	A small team can build and demonstrate locally while keeping an upgrade path (Compose and in-app JWT now).
003	Sarvam realtime speech-to-text over WebSocket; REST for short clips	Partial transcripts while the patient speaks; Indian-language accuracy; audio never retained.
004	Gemini with JSON-schema structured output, re-validated locally	Schema validity does not mean semantic safety, so every value is checked again; bounded calls, redaction, deterministic fallback.
005	Sarvam Bulbul TTS, Google TTS as a secondary adapter	Natural Indian voices; fixed UI audio pre-generated and cached; patient read-back synthesised in memory only.
006	Temporary CI waiver for one advisory in the CI-only Semgrep lock	Semgrep pins `mcp==1.29.0`; the app's own lock is audited with no exceptions.

14 Technology stack

Layer	Technology (version in the lock files)	Used for
Web	React 18.3, TypeScript 5.7, Vite 7	Single-page app for kiosk, phone and staff; route-level code splitting
	react-router 7, TanStack Query 5, TanStack Virtual 3, zustand 5	Routing; server state and caching; virtualised long lists (5,000-row queue); small client state
	i18next 24 + react-i18next 15	English, Hindi, Marathi in five namespaces; staff strings lazy-loaded
	Tailwind CSS 4, lucide-react icons, Inter and Noto Sans Devanagari (self-hosted)	Styling and type that renders Devanagari correctly offline
	vite-plugin-pwa 2 (auto-update)	Installable app, precached shell and prompt audio, offline short form
	react-hook-form, zod, qrcode.react	Forms and validation; token and check-in QR codes
API	Python 3.12, FastAPI 0.142, Pydantic 2.13, pydantic-settings	HTTP and WebSocket API, typed models, settings with production checks
	SQLAlchemy 2.1 (async) + asyncpg 0.31, Alembic 1.20	Database access (hand-written SQL in one repository module) and 22 reversible migrations
	Gunicorn 26 with Uvicorn workers	Process management (2 workers by default)
	redis-py 5.3, ARQ 0.28	Redis state, Lua scripts, pub/sub; the background worker
	argon2-cffi, PyJWT 2.15, cryptography 50	Argon2id passwords; HS256 tokens; AES-256-GCM and HMAC-SHA256
	google-genai 2.28, httpx, websockets 16, jsonschema 4.26	Gemini; Sarvam REST/WebSocket; validating every model answer
	WeasyPrint 70, Pillow 12, structlog 26, PyYAML	Case-sheet PDF; image checks and demo papers; structured logs; rules and weights files
Data	PostgreSQL 16.9, Redis 7.4.2 (two instances)	Records with row-level security; durable state and an evictable cache
Run	Docker Compose; nginx 1.27; optional Prometheus 3.5 + Grafana 12.1	One-command start (`start-demo.ps1` , `setup-hospital.ps1`); local metrics
Quality	pytest (+ coverage), Vitest 4, Playwright 1.63 + axe-core, Lighthouse CI	API unit and integration tests, web unit tests, browser tests with accessibility scans, performance budgets
	ruff, mypy --strict, ESLint 10, TypeScript, Prettier	Lint and types on every change
	Bandit, Semgrep, pip-audit, npm audit, Gitleaks	Security checks in CI
AI	Google Gemini (`gemini-3.5-flash-lite` , fallback `gemini-3.1-flash-lite` , TTS `gemini-3.8-flash-tts`); Sarvam (`saaras:v3-realtime` , `saaras:v4` clips, `bulbul:v3` voices)	Optional; selected by configuration ("auto" uses whichever keys are present)

15 Repository map

```
OPD-Sathi/
├─ apps/web/                       React app
│  ├─ src/app/                     screens: KioskIntake, KioskWelcome, KioskPapers, MobileCheckin,
│  │                               PatientStatus, StaffQueueBoard, DoctorConsultation, DocumentsPanel,
│  │                               PreviousVisits, AyurvedaPanel, NurseConsole, AnalyticsDashboard,
│  │                               AdminConsole, QueueSettingsCard, InterviewSettingsCard, router.tsx
│  ├─ src/lib/                     api client, staff auth, intake client, voice, offline outbox,
│  │                               photos, ids (secure-context safe), generated api-types.ts
│  ├─ src/locales/{en,hi,mr}/      common, kiosk, intake, status, staff strings
│  ├─ public/audio/                pre-recorded prompt audio per language
│  └─ e2e/                         Playwright specs (23 files)
├─ services/api/
│  ├─ app/main.py                  app factory, middleware, health, metrics
│  ├─ app/config.py                settings and production checks
│  ├─ app/api/v1/                  routers: auth, admin, audit, analytics, intake, audio, queue,
│  │                               case_sheets, documents, speech, ai_gate, visit_access
│  ├─ app/domain/                  interview (engine, bank JSON, consent texts), redflags (rules.yaml),
│  │                               triage (weights.yaml), routing, queue (state, scoring, tokens,
│  │                               Redis model), case_sheet, documents, patients, checkin
│  ├─ app/ai/                      summary, follow-ups, extraction, second opinion, triage review,
│  │                               department classifier, prescription, papers, speech, output guard,
│  │                               providers/{llm,stt,tts}
│  ├─ app/core/                    security, crypto, auth gate, rate limit, idempotency, audit, errors
│  ├─ app/db/                      session (tenant scope) and repositories (all SQL)
│  ├─ app/workers/main.py          outbox relay, notifications, analytics, close-out
│  ├─ app/cli.py, app/pdf.py       operator commands; case-sheet PDF
│  ├─ migrations/versions/         0001 … 0022
│  ├─ scripts/                     init-db, retention, schema/OpenAPI writers, demo papers
│  └─ tests/                       unit, integration (real PostgreSQL + Redis), perf
├─ eval/                           red-flag, routing, interview, extraction, adversarial sets
├─ infra/                          nginx config, observability, k6, chaos compose files
├─ scripts/                        backup.sh, restore.sh, lan-check.ps1, demo-check.ps1, chaos
├─ docs/                           plans, policies, privacy, evaluation, this documentation
├─ docker-compose.yml, Makefile, .env.example
└─ start-demo.ps1, setup-hospital.ps1
```

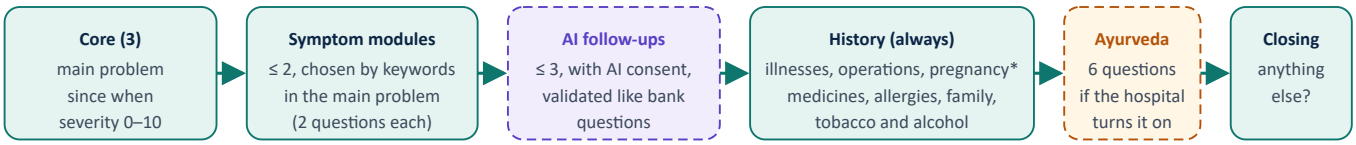
Where to look first: the interview starts in `domain/interview/engine.py`; urgency in `domain/redflags/engine.py` and `domain/triage/score.py`; the queue in `domain/queue/` and `api/v1/queue.py`; everything that touches the database in `db/repositories.py`; the kiosk in `apps/web/src/app/KioskIntake.tsx`.

16 Algorithms and rules

Every decision OPD Saathi makes about a patient comes from deterministic, readable rules in the `domain` package: no network, no database, no AI. AI output enters only as validated data and can only add caution. All rule files are marked `draft_needs_clinician_review`, and the loaders refuse them otherwise.

16.1 The interview engine

`services/api/app/domain/interview/engine.py` holds the interview as an immutable state (language, queued and asked question ids, answers, red-flag ids, AI questions, sex, age, extra modules). Every function returns a new state; the API stores it in the Redis session and every answer also as an encrypted `interview_turns` row.



Questions that cannot apply are skipped (sex, age, answers already given). Hard cap: 17 questions, plus 6 when the Ayurveda module is on.
* pregnancy only for women of the right age who did not already answer it.

Figure 16.1 The order in which questions are planned.

Planning the questions

1. **Core.** Every interview starts with `chief_concern` (11 tiles plus free text), `duration` (today, 1–3 days, 4–7 days, 1–4 weeks, more than a month) and `severity` (0–10 scale).
2. **Symptom modules.** When the main problem is answered, its text (NFKC-normalised, case-folded) is matched against keyword lists in `bank/triggers.json`; up to **two** modules are chosen, in bank order. Some phrases are ignored per module so they do not trigger wrongly (for example "fell ill" or "hair fall" never trigger the injury module).
3. **History, always:** past illnesses (with operations and pregnancy), medicines, allergies, family history, personal history (tobacco, alcohol), and the closing "anything else". A hospital's optional modules (Ayurveda) go just before the closing question.
4. **AI follow-ups** (only with AI consent and the hospital switch): when they arrive, up to 3 are inserted before the first unanswered history question, only if there is room under the cap; otherwise the engine records "skipped" and continues with the bank.

Module	Triggered by (examples; every language)	Questions
fever	fever, temperature, बुखार, ज्वर, ताप	fever pattern; associated symptoms (rash…)
respiratory	breath, cough, wheez, phlegm, साँस, खाँसी, खोकला, धाप, कफ	breathing symptoms (blood in phlegm, breathless, wheeze…); TB contact, weight loss
digestive	stomach, abdomen, vomit, loose motion, पेट, उल्टी, पोट, जुलाब	stomach symptoms; where it hurts
urinary	urine, urinat, पेशाब, लघवी	urine symptoms; associated (flank pain, swelling)
pain	pain, ache, sore, headache, दर्द, वेदना, दुखत	where; what it is like (spreading…)
skin	skin, rash, itch, त्वचा, खुजली, पुरळ	skin problem; where
mobility (injury)	injur, fall, accident, fracture, sprain, bite, चोट, अपघात	what happened (road accident, animal bite…); effect (cannot walk, wound…)

Module	Triggered by (examples; every language)	Questions
sleep_energy	tired, weak, fatigue, sleep, dizz, थकान, चक्कर, थकवा	tiredness, dizziness, weight loss, night sweats

Which questions a patient actually sees

A question (and each of its options) can declare who it applies to: `only_sexes`, `min_age` / `max_age`, `hide_with_modules` (skip when another module covers it) and `hide_with_answers` (skip when an earlier answer makes it pointless). A question whose real options are all hidden is skipped; one listed under `skip_if_only_concerns` is skipped when the patient only tapped those concerns. The progress bar shows answered + remaining, never more than the cap.

Answers

- **Validation:** a scale answer must be a whole number 0–10; text has control characters removed, whitespace collapsed and at most 500 characters; required questions cannot be empty; optional ones can be skipped. An answer must be for the current question, otherwise it is refused (the API first checks whether it already saved that exact answer, which makes retries safe).
- **Back** removes the last answer and returns it for pre-filling; going back to the main problem forgets the symptom questions and AI follow-ups planned from it.
- **Correction** on the review screen replaces an answer. If a changed main problem plans different modules, answers that no longer fit are dropped and new questions become pending; AI follow-ups written for the old problem go too.
- Answers are fixed once the database records a token; the check is against the database, not the session.

From answers to the case sheet

Answer	Case-sheet field
chief_concern, duration, severity	chief complaint; HPI onset; HPI severity (0–10)
pain / digestive site / skin site	HPI location (first one answered)
pain character / fever pattern	HPI character
other symptom questions, AI follow-ups	HPI associated symptoms (up to 30)
past_history, surgeries, pregnancy	past conditions, operations, women's health
medications, allergies, family_history	medicines (patient-stated), allergies, family history
habits	social: tobacco (smoking, chewing), alcohol, other habits in the patient's words
Ayurveda answers	`ayush`: appetite, digestion, bowel, sleep, exercise, diet
"None of these"	a **pertinent negative** listing what was offered: "Denies vomiting, loose motions, …", "No family history of diabetes, high BP, …", "No known drug or food allergy", "No previous operations"
"Don't know", "Prefer not to say"	**unanswered**: "Does not know: …" / "Preferred not to say: …", never a denial

Each text field keeps the patient's original words and an English version: a tapped option uses its bank English label (confidence 0.9), so the doctor gets English even without AI; typed or spoken words stay as said (0.5) until AI translation, which is marked "AI-assisted". The sheet is validated against the strict, versioned `CaseSheet` schema (`domain/case_sheet.py`, schema 1.0; CI fails if the published JSON schema drifts).

16.2 The red-flag engine

`domain/redflags/engine.py` evaluates 15 rules from `rules.yaml` against the patient's text and facts after every answer. Results are deduplicated, ordered and stored once per visit and rule (`red_flag_events`); an urgent flag raises a

staff alert at once.

Matching, step by step

1. **Normalise** the text: Unicode NFKC and case-folding; Devanagari nukta removed (so precomposed and decomposed spellings match); chandrabindu treated as anusvara (common informal typing); punctuation becomes spaces; whitespace collapsed. Then split into tokens.
2. **Check the rule's structured conditions** against the patient's facts: `tile_any` (a danger tile tapped), `min_age` / `max_age`, `sex_any`, `pregnant`, `severity_min`. A rule whose conditions fail is skipped.
3. **Match phrase groups.** A rule has one or more groups; each group lists phrases in English, Hindi, Marathi and Hinglish (the loader refuses a group missing a language). **Every** group must match (AND); within a group **any** phrase may match (OR). A phrase matches as a whole token sequence.
4. **Negation.** A match is ignored when a negation word (no, not, without, nahi, nahin, nhi, नहीं, नही, नाही…) appears in the 3 tokens before it, or in the 3 tokens after it unless a contrast (but, however, लेकिन, पण…) or conjunction (and, और, आणि, aur, ani) comes between.
5. **Evidence.** The flag records the tapped tile, or two words either side of the earliest match (at most 240 characters), so staff read exactly what triggered it.

Example: negation

"I have **no chest pain**" → the phrase "chest pain" has "no" within 3 tokens before it → no flag.

Example: two groups

"chest pain, not sweating" → group 1 ("chest pain") matches, group 2 ("sweating") is negated → `RF_CHEST_PAIN_ASSOC` (urgent) does not fire; `RF_CHEST_PAIN` (priority) does.

The 15 draft rules

Rule	Level	Conditions	Phrases (English examples; every rule also has Hindi, Marathi and Hinglish)
`RF_DS_TILE`	urgent	any danger tile	(tile only) chest, breathing, bleeding, fainting/fits, stroke signs…
`RF_CHEST_PAIN_ASSOC`	urgent	–	chest pain / pressure / tightness **and** sweating / breathless / left arm pain
`RF_CHEST_PAIN`	priority	–	chest pain, chest pressure, chest tightness
`RF_BREATH_SEVERE`	urgent	–	cannot breathe, cannot speak full sentences, severe breathlessness, blue lips
`RF_STROKE_SIGNS`	urgent	–	face drooping, one-sided weakness, slurred speech, sudden vision loss
`RF_BLEEDING_HEAVY`	urgent	–	heavy bleeding, vomiting blood, blood in stool
`RF_PREGNANCY_DANGER`	urgent	pregnant	vaginal bleeding, severe abdominal pain, fits, baby not moving
`RF_CHILD_FEVER`	urgent	age < 1	fever, high temperature
`RF_CHILD_FEVER_ASSOC`	urgent	age ≤ 4	fever **and** fits / seizure / not feeding / very sleepy
`RF_SEIZURE_FAINT`	urgent	–	seizure, fits, unconscious, fainted
`RF_SELF_HARM`	urgent	hidden from patient	want to die, kill myself, harm myself, suicidal thoughts (private staff follow-up)
`RF_TRAUMA_MAJOR`	urgent	–	road accident, fell from height, head injury with vomiting, deep wound
`RF_ALLERGY_SEVERE`	urgent	–	throat or face swelling with breathing trouble, hives with breathing trouble
`RF_POISON_BITE`	urgent	–	poison swallowed, overdose, snake bite, scorpion sting

Rule	Level	Conditions	Phrases (English examples; every rule also has Hindi, Marathi and Hinglish)
`RF_HIGH_FEVER`	priority	–	fever 104, temperature 104, very high fever

Tapped answers count the same in every language. Three tapped options raise a rule directly, whatever the language of the label: blood in stool → `RF_BLEEDING_HEAVY`, very high fever → `RF_HIGH_FEVER`, road accident → `RF_TRAUMA_MAJOR` (`option_flags` in `weights.yaml`).

> **The AI second opinion can only add caution**
>
> With AI consent, a bounded model call (5 s) reviews the interview against the same rule list. It can only preserve or raise urgency, must quote the patient's words, and its flags count as **priority at most** in the score until a nurse confirms ("AI suggests, nurse to confirm"). In one live test the model matched "chest pain with sweating" from "breathless on walking"; this is why AI is a checker, not a decider.

16.3 The triage score

`domain/triage/score.py` adds up explainable points (0–100) from things that are hard to exaggerate, and only lightly from the patient's own rating. Every point is stored with its reason (the **Why?** panel). Weights live in `weights.yaml` (version 1, draft).

Component	Points
Red flags	The first flag: urgent **70**, priority **30** (an AI flag counts as priority). Each further flag **+5**, at most **+15** together.
Warning signs (tapped answers)	Blood in phlegm 25 · difficulty breathing as main problem 12 · blood in urine 12 · animal bite 12 · breathless on walking 10 · very little urine 10 · bleeding or open wound 10 · lower-right abdominal pain 8 · swelling of feet or face 8 · cannot walk since injury 8 · injury or fall 6 · wheezing 6 · flank pain 6 · fever with rash 6 · wound not healing 6 · known heart disease 6 · TB at home 5 · weight loss 5/4 · spreading pain 5 · dizziness 5 · vomiting 4 · loose motions 4 · fever with shivering 4 · swelling after injury 4 · known diabetes 3 · night sweats 3. **All together at most 25.**
Vulnerability	Infant under 1: 20 · child 1–4: 10 · aged 75+: 12 · aged 65–74: 8 · pregnant: 12
Onset	Started today: 6 · 1–3 days ago: 3
Own rating (0–10)	7 → 1 · 8 → 2 · 9 → 3 · 10 → 4. None for a problem lasting more than a month.

```
score = min(100, sum of points)
level = urgent if any urgent flag (not counting AI flags) or score ≥ 70
        priority if score ≥ 30
        routine otherwise
```

Worked examples

Patient	Points	Score	Level
60 y man, taps **Chest discomfort**, started today	danger tile 70 + today 6	76	urgent
78 y man, main problem difficulty breathing, breathless on walking, started today, rates 7	12 + 10 + aged 75+ 12 + today 6 + rating 1	41	priority
34 y woman, fever with shivering for 1–3 days, rates 8	4 + 3 + 2	9	routine
An adult who taps every warning sign that is not itself a red flag, started today, rates 10	signals capped at 25 + today 6 + rating 4	35	priority, never urgent

Monotonic urgency

The score is recomputed after every answer, but a visit's urgency never falls on its own: only a nurse or administrator can lower it, with a written reason that goes to the audit log. A self-reported danger sign keeps the patient first and marked "Nurse to confirm" until acknowledged.

With AI consent, a **triage review** (12 s, after confirmation) reads the whole interview and says whether the level fits. "More urgent" raises a priority alert for a nurse; "less urgent" is a note; a failure leaves the score alone. Its verdict never moves the queue by itself.

16.4 Calling order and the head start

`domain/queue/scoring.py` turns the level and score into a **virtual arrival time**; lower is called first.

```
head_start (minutes) = min(45, round(score × 0.6))
  routine: capped at 17 (just below the priority floor of round(30 × 0.6) = 18)
  priority: at least 18
virtual_arrival = queued_at − head_start
calling order = (urgent first, virtual_arrival, visit id)
```

- **Urgent patients are always first**, however long others have waited.
- Bounding the head start by level means a nurse's change of level always wins over the points.
- A routine patient can only be overtaken by someone who arrived less than 45 minutes after them: nobody waits for ever.
- `queued_at` is kept through a recall, so a recalled patient returns to their original place. Ties are broken by visit id, which is a time-ordered UUIDv7. Visits from before the triage score existed use fixed bonuses (urgent 360, priority 45, vulnerable +15 minutes).

Patient	Arrived	Level, score	Head start	Virtual arrival	Called
A	10:00	routine, 9	5 min	09:55	3rd
B	10:15	priority, 41	25 min	09:50	2nd
C	10:30	urgent, 76	–	–	1st
D	10:05	routine, 0	0 min	10:05	4th

Call next runs this order on the server inside a department-wide transaction lock, so two doctors pressing it together get two different patients, and a doctor already holding a patient (called or in consultation) is refused with `DOCTOR_BUSY`.

16.5 The wait estimate

```
consult = average of the last 20 consultations once there are 5, else 4 minutes; kept within
1–15
baseline = (position − 1) × consult ÷ max(1, active doctors)
shown range = [ceil(0.75 × baseline ÷ 5) × 5 , ceil(1.25 × baseline ÷ 5) × 5] minutes
```

Example: 5th in line, 2 doctors, 6-minute consultations → baseline 12 → "10–15 min". A five-minute range avoids a falsely precise promise.

16.6 Department routing

`domain/routing/engine.py` chooses a department from the patient's stated concern by keyword modules, and sends anything unclear to a nurse.

1. **Surgery first:** "abdominal lump", "stomach lump", "hernia" (and Hindi/Marathi equivalents) → `SURG`, confidence 0.90.
2. **Match every module** (ORTHO, ENT, OBG, DERM, OPHTH, SURG, GENMED) on the normalised text; each module contributes its earliest matching phrase. `OBG` counts only for a woman aged 12–50 or a pregnant patient.
3. **Children:** under 14 with any match → `PAED`, 0.90.
4. **No match** → `NURSE_TRIAGE`, 0.0, needs nurse confirmation.
5. Otherwise the module whose phrase appears **earliest** in the text wins: confidence 0.90 if only one department matched, 0.65 if several did. Below 0.70 the queue row shows "Department not confirmed" for a nurse.

Said	Result
"joint pain since a week"	ORTHO, 0.90
"ear pain with fever"	ENT (earliest) and GENMED matched → ENT, 0.65, nurse to confirm
"stomach lump"	SURG, 0.90 (surgery rule first)
8-year-old, "cough"	PAED, 0.90
"feeling tired"	NURSE_TRIAGE, nurse to confirm

Kiosks start every patient in General Medicine; confirmation applies the route. With AI consent a **department classifier** (5 s, configured labels only) can suggest a department; disagreement with the rules goes to a nurse. Nurse confirmation (`POST /queue/{id}/routing/confirm`) and overrides are audited. Evaluation: 224/224 dev and 96/96 test cases (synthetic, section 21).

16.7 The queue state machine

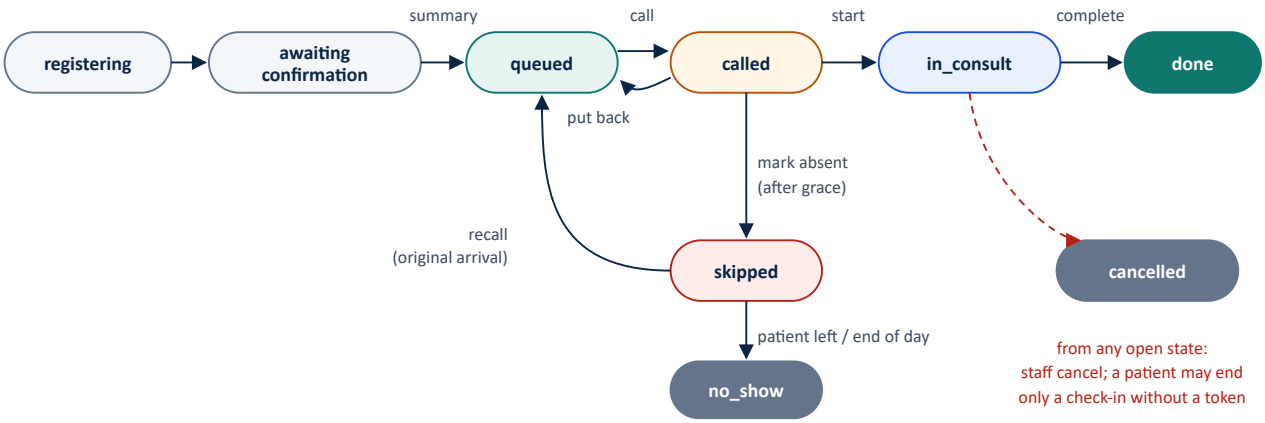


Figure 16.2 Visit states (`domain/queue/state.py`). `done` , `no_show` and `cancelled` are final.

From	Allowed to
registering	awaiting_confirmation, cancelled
awaiting_confirmation	queued, cancelled
queued	called, cancelled
called	in_consult, skipped (after the grace period), queued (put back), cancelled
in_consult	done, cancelled

From	Allowed to
skipped	queued (recall), no_show, cancelled
done, no_show, cancelled	nothing (final)

- **One active claim per doctor:** a doctor may hold at most one visit in `called` or `in_consult`; `called_by` records who called. A doctor cannot act on another doctor's patient (`CLAIMED_BY_ANOTHER_DOCTOR`); a nurse or admin can mark that patient absent or put them back.
- **Grace period** (default 180 s, 0–900 per hospital) is enforced by the server before `skip`.
- Every transition takes the visit row lock, writes an append-only `visit_events` row (from, to, actor type and person, reason) and an audit record, and an outbox event for the live screens.

16.8 Tokens, status links, OPD cards and check-in codes

Value	How it is made	Why
Token `MED-007`	Department prefix + a daily number per hospital, department and IST date, at least three digits. Allocated under a PostgreSQL advisory lock as max(today's numbers) + 1, with a Redis counter as the fast path (a Redis outage falls back to the database).	Human-friendly, gap-free and unique even with many kiosks confirming at once.
Status link `/s/<43 chars>`	HMAC-SHA256(APP_HMAC_KEY, "osa:status:v1:{hospital}:{visit}") in URL-safe base64 (256 bits). Only its SHA-256 digest is stored.	Unguessable, re-printable, reveals nothing, cannot be enumerated.
OPD number `26-000123`	Year + a per-hospital, per-year counter (`patient_number_counters`), assigned at the first confirmed visit.	The hospital registration number on the card.
Card PIN (6 digits)	First 8 bytes of HMAC-SHA256(APP_HMAC_KEY, "osa:card-pin:v1:{hospital}:{patient}:{card_version}") mod 1,000,000. Compared in constant time. Never stored.	Proves the patient holds the card; a lost card is replaced by moving to the next version.
Phone check-in code	HMAC-SHA256 over hospital, check-in point and version (URL-safe, 43 characters); only a SHA-256 digest is stored for lookup.	The QR can be shown again any time and replaced at once.
Kiosk secrets	One-time activation code and the device secret: 256-bit random values stored as SHA-256 digests.	A stolen database does not reveal live credentials.

16.9 The AI pipeline

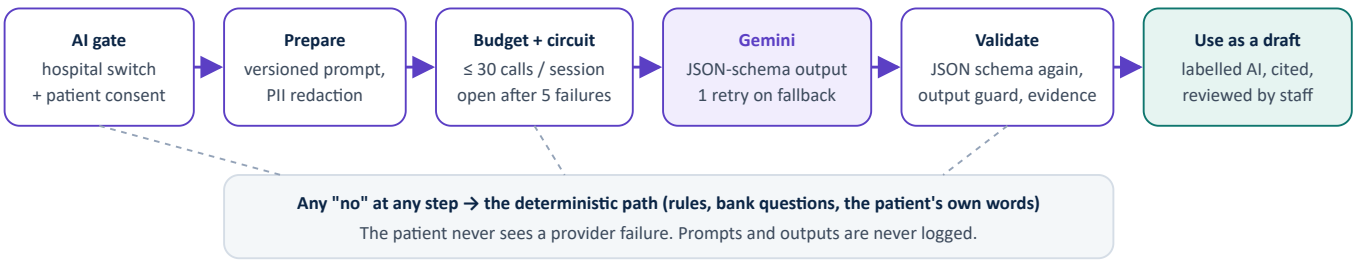


Figure 16.3 Every language-model call goes through the same pipeline (`ai/providers/llm/gemini.py`).

- **Deadline and retry:** each call has a total deadline (callers set it; at most 20 s). The first attempt gets 65% of it; a retryable failure (timeout, connection error, HTTP 408, 429 or 5xx) retries once on the fallback model after 0.1–0.3 s of jitter.
- **Circuit breaker:** 5 consecutive failures open it for 20 s; then exactly one probe is allowed. Text-to-speech has its own failure handling so a voice outage never blocks text features.
- **Budget:** a Redis counter per session (hashed id) allows 30 calls in 2 hours, shared across API workers. Output is capped at 1,024 tokens; "thinking" is set to minimal (stepping down if a model refuses it).
- **Validation:** the response must parse as JSON and match the schema (Draft 2020-12) again locally; the output guard then checks every string; feature code adds its own evidence checks (citations limited to asked questions, red flags must quote the patient, follow-up questions must be trilingual and safe).

Feature (module)	Time limit	Output and checks
Follow-up questions (`followup_questions.py`)	background	≤ 3 questions in en/hi/mr with tap options; unsafe or malformed ones dropped; bank used otherwise
Clinical summary (`clinical_summary.py`)	15 s	English fields + brief as cited sentences; original words never replaced; failure keeps the deterministic sheet
Red-flag second opinion (`redflag_opinion.py`)	5 s	Only rule ids from the list; must quote; priority at most until confirmed
Triage review (`triage_review.py`)	12 s	agree / more urgent (alert) / less urgent (note)
Department classifier	5 s	A configured label; disagreement → nurse
Question choice (off by default)	1.5 s	Picks among configured question ids only
Prescription reading (`prescription.py`)	15 s	Medicine names; patient confirms; photo not stored
Paper reading (`document_reading.py`)	20 s	Draft reading stored encrypted; doctor accepts or rejects
Speech to text (`speech.py`)	12 s	Verbatim in the patient's script; Sarvam → Gemini → browser
Extraction (`extraction.py`)	–	Allow-listed case-sheet fields from voice turns, each checked against evidence

Confirmation never waits long for AI: in the scenario tests a hanging AI service adds at most about 5 seconds to confirmation, and a failed one adds nothing visible.

16.10 Old papers: reading and printed-range flags

- **Accepted files:** JPEG, PNG or WebP whose first bytes match the declared type, at most 1,000,000 bytes (the kiosk sends a ~1280 px JPEG of about 200 kB). Up to 5 per check-in by the patient, 10 per visit in all. Each photo is encrypted with AES-256-GCM bound to its document id, so a photo copied onto another record cannot be decrypted.
- **Reading** runs in the background (25 s budget) only with AI consent; a reading still pending after 2 minutes is shown as failed. The model is told to copy what is clearly written and not to interpret.
- **Flags by arithmetic** (`domain/documents.py`):

```
value : a number, optionally followed by its unit ("9.8 g/dL", "2,10,000 /cumm")
range : "12 - 15", "12–15", "12 to 15", "(4,000 - 11,000)", "< 5.7", "up to 5", "> 40"
digit grouping: "1,50,000" and "150,000" are both 150000
flag : above if value > high · below if value < low · within otherwise (a value at a limit is
within)
no flag: text results ("No growth"), censored values ("< 0.5"), several ranges, reversed
ranges, anything unparsed
```

A high or low mark printed by the laboratory ("H", "LOW", ↑, ↓) is shown as printed, separately. No general reference range is ever applied, because ranges differ between laboratories, methods, ages and sexes. A printed date is used only if it is a real date between 1950 and today.

16.11 Notifications

- After every relayed queue change, the worker looks at the first six patients and records one-time intents: `token_issued`, and `five_ahead`, `two_ahead`, `your_turn` when a patient reaches position 6, 3 and 1. Only visits with SMS opt-in and a phone get intents, deduplicated per visit and template.
- At most 5 messages per recipient per hour (serialised with an advisory lock on the phone digest). Delivery retries after 1, 5 and 30 seconds, then the message is marked failed. Messages carry only the token and position, in the patient's language, and the phone number is never logged.
- Providers: `console` (default, logs only), Twilio SMS, WhatsApp Cloud API.

> **Known defect (Phase 7 task 7.0.1)**
>
> The worker deliberately has no patient-data key, yet delivery decrypts the phone number in the worker. On the demo stack this makes the notification step fail every 2 seconds once a patient has opted in; queue updates still work because they commit first. The fix gives the worker a separate, narrower key for opted-in phone numbers only.

16.12 Reports

A materialized view, `osa_analytics.analytics_daily_department`, rolls visits up per hospital, department and day: visits, urgent visits, total waiting, consultation and interview time with their sample counts, confirmed case sheets, those confirmed without edits, doctor-edited ones, visits with a case sheet, and the language mix. The worker refreshes it concurrently every 5 minutes under a dedicated role (`osa_analytics_worker`); the API filters it by hospital and computes the cards (means are rounded to 0.1). Measured: the summary query takes 1.88 ms p95 on 100,000 synthetic visits.

16.13 Rate limits, idempotency and session locks

Mechanism	How it works
Sliding-window limits	A Lua script on Redis keeps a sorted set of request times per key, trims the window, and answers allowed / remaining / retry-after (`Retry-After`, `X-RateLimit-*` headers). The sign-in limit **fails closed** if Redis is down; the others fall back to a bounded in-process window. Examples: sign-in 10/min; phone check-in 6 per phone per QR and 150 per QR per 10 min; OPD card 5 per visit per 15 min and 10 per number per hour; "Call staff" 5 per visit per 10 min; paper photos 10 per patient and 40 per staff member per 10 min.
Idempotency	Every staff write requires an `Idempotency-Key` (1–200 visible characters). The key is stored per hospital, actor, method and path with a hash of the body: a repeat returns the stored response for 24 hours; the same key with a different body is refused (422); a concurrent duplicate waits up to 5 s. Ownership checks stop an expired reservation from overwriting a retry.
Session lock	One change at a time per patient session (Redis lock, 20 s), so a double tap cannot store two answers; a re-sent answer that was already saved returns where the interview stands.
Optimistic concurrency	Case-sheet edits and AI reviews send `If-Match` with the version (`ETag`); a stale version gets 409 and the screen offers reload or re-apply.

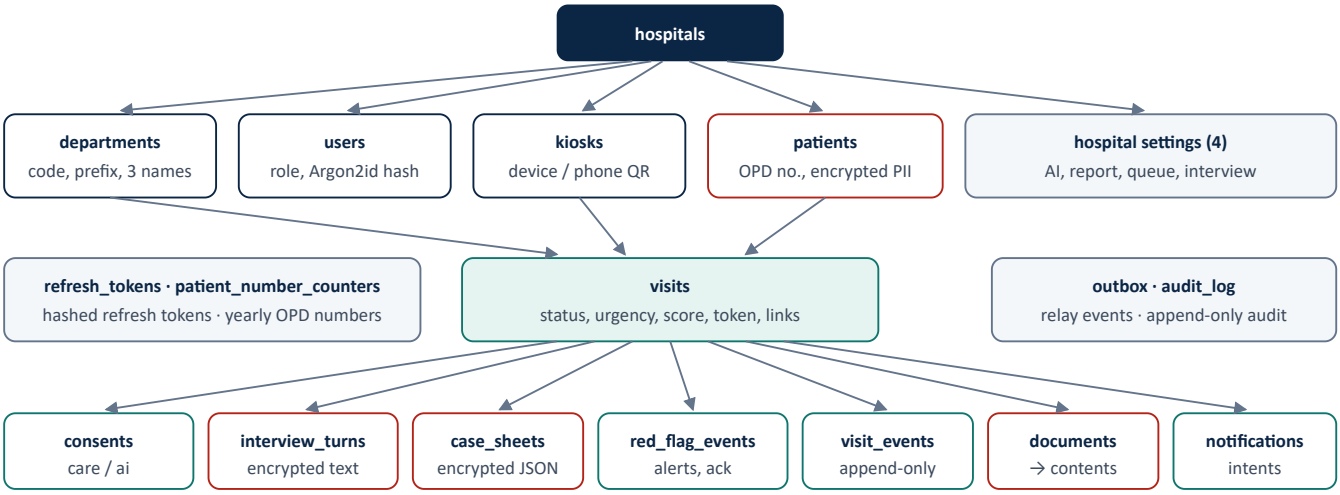
16.14 End-of-day close-out

At minutes 1, 16, 31 and 46 of every hour (and at API and worker start-up) visits left open on an earlier IST day are closed, each with a `day_closed` timeline entry, and the Redis queues are rebuilt:

Open state	Becomes	Why
queued	cancelled	Not seen because the OPD closed; not the patient's absence
called, skipped	no_show	Called and never seen
in_consult	done	The consultation was not closed on screen
registering, awaiting_confirmation	cancelled	No token was issued

17 Data model

22 tables and one materialized view in PostgreSQL, created by 22 reversible Alembic migrations. Every table that holds hospital data carries `hospital_id`, is protected by row-level security, and references other tables through composite `(hospital_id, id)` foreign keys, so a row can never point into another hospital.



Red outline: holds encrypted fields. Every table carries hospital_id and is filtered by row-level security.

Figure 17.1 The main tables and how they relate.

Table	What it holds	Protection
`hospitals`	Tenant root: name, unique short code	Names readable by the worker for the sign-in picker
`departments`	Code, token prefix, names in three languages, active flag, archive	RLS
`users`	Staff: role, name, email, password hash, department, failed sign-ins, lock time, active, archive	RLS; Argon2id
`refresh_tokens`	Per-account refresh tokens (rotated)	SHA-256 digest only
`kiosks`	Device kiosks and phone check-in points: label, kind, activation code hash and expiry, device secret hash, check-in code hash and version, last seen, paired at	RLS; secret lookups through narrow policies before the tenant is known
`patients`	Per-hospital person: OPD number (`mrn`), card version, name, phone and ABHA (encrypted), phone digest (not unique), birth year, sex, language	RLS; AES-GCM; HMAC digest
`patient_number_counters`	Next OPD number per hospital and year	RLS

Table	What it holds	Protection
`visits`	The encounter: patient, department, kiosk, channel, service date, status, urgency, acuity score and breakdown, token number and label, status-token hash, queued/called/skipped/recalled times, `called_by`, skip count, SMS opt-in, patient link (`card` / `staff`), previous visit	RLS; composite FKs
`consents`	One row per purpose (`care`, `ai`): version, text SHA-256, language, answer, channel, time, withdrawal (when, by patient or staff)	RLS; audit
`interview_turns`	Every question and answer in order: question id, input mode, original text, English text, speech confidence, corrections	RLS; AES-GCM per turn
`case_sheets`	The structured history as encrypted JSON, version, patient confirmation, last editor	RLS; AES-GCM; If-Match versions
`red_flag_events`	One row per visit and rule: level, source, evidence, acknowledged by and when	RLS
`visit_events`	The queue timeline: from, to, actor type and person, reason, time	RLS; insert/select only
`documents`	Old papers: kind, source (patient/staff), uploader, printed date, reading status, encrypted reading, review status, reviewer and time	RLS; AES-GCM
`document_contents`	The photo bytes, cascading with their document	RLS; AES-GCM bound to the document
`notifications`	Message intents: template, channel, status, attempts, next attempt, recipient digest, dedupe key (never the phone number)	RLS
`outbox`	Queue events for the worker relay, processed time, attempts	RLS
`audit_log`	Who did what to which record, when, with safe metadata	RLS; append-only for the app role
`hospital_ai_settings`, `hospital_report_settings`, `hospital_queue_settings`, `hospital_interview_settings`	AI switch; minutes saved per visit; call grace period; Ayurveda module	RLS; changes audited
`osa_analytics.analytics_daily_department`	Materialized daily rollup for reports	Separate schema; filtered by hospital through a security-barrier view

Migrations

Migration	What it added
0001 hospitals	The tenant root table
0002 identity_security	Users, kiosks, refresh tokens, append-only audit
0003 secret_lookup_rls	Exact high-entropy secret lookups before the tenant is known
0004 domain_data	Departments, patients, visits, consents, case sheets, interview turns, red flags, outbox, notifications, documents
0005 queue	Queue read-path indexes and status-token lookup
0006 redflag_events	One durable red-flag event per visit and rule
0007 hospital_ai_settings	Per-hospital AI opt-in switch
0008 patient_assistance	The patient-help flag for the staff queue
0009 / 0010 notification_*	Per-visit SMS opt-in; idempotency, retries and recipient limits
0011 analytics_rollup	The daily department rollup and its refresh role
0012 hospital_directory	Hospital picker reads; worker close-out of previous days
0013 triage_score	Explainable triage score and breakdown per visit
0014 analytics_case_sheets	Case-sheet counts in the rollup
0015 report_settings	Minutes saved per visit (time-saved card)
0016 kiosk_lifecycle_and_archive	Kiosk pairing lifecycle, archive (delete) for admin records, visit kiosk attribution
0017 mobile_checkin	Phone check-in points (`kiosks.kind` , check-in code hash)
0018 patient_identity	OPD number and card version, shared phones (non-unique digest), visit links and follow-ups
0019 queue_recall	`skipped` , `called_by` , recall fields, `visit_events` , grace period
0020 interview_settings	Optional Ayurveda module per hospital
0021 consent_purposes	Consent purpose (care/ai) and withdrawal
0022 documents	Old papers: kind, source, review, encrypted photo table

CI applies every migration up, down to base and up again on a real PostgreSQL. A lesson recorded in the build log: a data step in a downgrade must lift `FORCE ROW LEVEL SECURITY` first, or it silently touches no rows.

Database roles

Role	Used by and allowed to
`osa_app`	The API. No superuser, no DDL; every query runs inside a transaction whose hospital scope is set with `set_config('app.hospital_id', …, true)` ; cannot update or delete audit and timeline rows.
`osa_migrator`	Alembic migrations only: owns the schema.
`osa_worker`	The worker: relays every hospital's queue, so it can read across hospitals, but has only the table grants its jobs need and no patient-data key.
`osa_analytics_worker`	No login; assumed by the worker only to refresh the analytics view.

18 API

96 routes (94 HTTP, 2 WebSocket), all under `/api` through nginx. The full list is in Appendix A; the OpenAPI document is served at `/openapi.json` and `/docs` outside production, and TypeScript types are generated from it (CI fails if they drift).

Conventions

Topic	Rule
Authentication	`Authorization: Bearer <token>` on everything except an explicit allow-list: health and readiness, the hospital list, sign-in/refresh/sign-out, kiosk activation and token renewal, the status page (and its stream and "stop AI help"), and the phone check-in lookup and start. Public routes that carry a code accept only the exact 43-character URL-safe shape. WebSockets authenticate in the first frame.
Roles	Five token roles: `admin`, `doctor`, `nurse`, `kiosk`, `patient`. Each route declares the roles it accepts; visit routes also check department and, for patients, that the visit is their own.
Errors	RFC 7807 `application/problem+json` with a stable machine `code` and the `request_id`, for example `{"title": "…", "status": 409, "detail": "…", "code": "DOCTOR_BUSY", "request_id": "…"}`. Records the caller may not see answer "not found".
Writes	Staff writes require `Idempotency-Key`; case-sheet edits and AI reviews require `If-Match` with the version from `ETag`.
Lists	Keyset pagination with `limit` and an opaque `next_cursor` signed with `CURSOR_HMAC_KEY` (tampered cursors are refused).
Tracing	Every response carries `X-Request-ID` (the caller's, if valid, else a new UUID); logs carry the same id.
Limits	Request bodies over 1 MiB are refused with 413 before parsing; rate-limited routes return 429 with `Retry-After`.

Route groups

Group	Routes	Highlights
Auth and kiosks	7	`/auth/hospitals`, `/auth/login`, `/auth/refresh`, `/auth/logout`, `/kiosks/activate`, `/kiosks/token`, `/kiosks/heartbeat`
Intake (patient session)	22	sessions, consent and AI consent, identity, OPD card, follow-up, danger tiles and signs, next question, answers, correction, back, language, help, needs-assistance, read-back audio, confirm, end session, offline submissions, phone check-in, assisted sessions
Speech	4	capabilities, transcribe, speak (approved text only), prescription photo
Queue, alerts, status	21	queue list, call-next, call/start/complete/skip/recall/return/no-show/cancel/override, routing confirm, link patient, patient lookup, register, alerts and acknowledge, status, status stream, stop AI help, visit timeline
Case sheets	7	read, patch, PDF, transcript, history, AI review, AI-consent withdrawal by staff
Documents	7	patient add/list/remove; staff list/add, photo content, review
Admin	17	generic list/create/patch/delete for users, kiosks, departments; activation codes; check-in QR show/replace; AI, queue and interview settings
Analytics, audit	6	summary, timeseries, CSV export, report settings; cursor-paginated audit log
Operations	3	`/healthz`, `/readyz` (PostgreSQL and Redis within 1 s), `/metrics` (404 unless enabled)
WebSockets	2	`/ws/staff/queue`, `/ws/intake/{visit}/audio`

19 Security and privacy

Health data was the design constraint from the first commit. Security is layered so that one mistake does not expose patients: the database refuses cross-hospital rows even if the application is wrong, encrypted fields are useless without the key, and every important action is recorded.

19.1 The layers

Layer	Controls in place
Network	Only the web port is published; PostgreSQL, both Redis instances and the API listen on 127.0.0.1 or the private Compose network; Redis requires a password. nginx overwrites (never appends) the forwarded address.
Browser	Strict Content Security Policy (scripts only from the app's own origin, no inline script, no framing), `X-Frame-Options: DENY`, `nosniff`, `Referrer-Policy: no-referrer`, microphone and camera only for the app. API responses: `default-src 'none'`; HSTS in production.
Identity	Argon2id passwords (12–1,024 characters); lockout after 5 failures for 15 minutes; sign-in rate limit that fails closed; HS256 tokens with key id, issuer, audience, unique id and expiry, checked strictly (30 s clock leeway); refresh tokens and kiosk secrets stored only as SHA-256 digests; one-time 24-hour kiosk activation codes.
Authorisation	An explicit public-route allow-list in middleware; per-route roles; department checks on every visit route and queue action; the earlier-records rule (open or today's visit); a patient session reaches only its own visit and can change it only until the token is issued.
Tenant isolation	PostgreSQL row-level security, forced, on every tenant table, scoped per transaction; composite tenant foreign keys; secret lookups before the tenant is known go through narrow, exact-match policies.
Data at rest	AES-256-GCM for names, phones, ABHA numbers, interview text, case sheets, paper readings and photos, each bound by associated data to its table, column and row (`patients:phone_enc:<id>`), so ciphertext moved to another row fails to decrypt. Keyed HMAC-SHA256 digests for lookups. The kiosk's offline form is AES-GCM encrypted under a non-extractable browser key.
Minimisation	Only age and sex required; audio never stored; prescription-scan photos processed in memory; patient read-back audio synthesised in memory with `no-store`; after confirmation the Redis session keeps only visit, kiosk, channel and language; logs carry no answers, phones, tokens or codes; metrics use route templates, never ids.
Integrity	Idempotent writes; optimistic versions; append-only audit log and visit timeline (no update or delete grant for the app); transactional outbox.
Abuse	Rate limits on every public and patient route; 1 MiB request limit; strict request models that reject unknown fields; file-signature checks on photos; "Call staff" and phone check-in limits.
Configuration	In production the settings refuse development signing keys, short secrets and malformed 32-byte keys, enforce the 10-minute access-token lifetime and hide the API docs. The setup scripts generate random secrets and never reuse published defaults.
Supply chain	Python dependencies installed with `--require-hashes` from a lock file; `npm ci`; pip-audit, npm audit, Bandit, Semgrep (0 findings) and Gitleaks on every change; the API image runs as a non-root user; an SBOM on request (`make sbom`).

19.2 Tokens and their lifetimes

Credential	Lifetime	Stored as
Staff access token (JWT)	10 minutes (enforced)	Memory only
Staff refresh cookie	8 hours, rotated	HttpOnly, SameSite=Strict, Secure outside development; server keeps a digest

Credential	Lifetime	Stored as
Kiosk activation code	24 hours, single use	Digest
Kiosk device secret → kiosk token	Until re-paired or deleted → 12 hours	Browser storage on the kiosk; server keeps a digest
Patient session token	2 hours	Kiosk: tab storage; phone: local storage until the day ends
Status link, check-in code, OPD PIN	Stable (derived)	Digest (status, check-in) or not at all (PIN)

19.3 Consent and the patient's control

- **Care** consent is required; **AI** consent is optional and is asked separately with nothing pre-selected. Each is stored with the notice's version and text hash, language, channel and time.
- The AI answer can be changed until confirmation and withdrawn afterwards by the patient (status page) or by staff. Withdrawal stops all further AI processing, discards work still running, hides the AI draft and paper readings, and blocks their review. Processing done before the withdrawal is not undone.
- Without AI consent, nothing about the visit leaves the server: no model call, no online speech recognition, no browser speech recognition. This is checked in tests by counting calls that reach a stub provider.

19.4 What is audited

Sign-in success, failure and lockout; every queue action (`queue.called` , `queue.skipped` , `queue.recalled` , `queue.override` …); alert acknowledgements; case-sheet reads and edits; AI reviews; history reads (`patient_history.read`); patient lookups and links; paper list views, photo views, additions, removals and reviews; consent answers and withdrawals; OPD-card attempts (outcome only); every administrator change; report views. Audit metadata never contains patient answers or credentials.

19.5 How isolation is tested

`tests/integration/test_access_isolation.py` keeps an inventory of every route that names a record. Each is probed as another hospital's staff, another department's staff and another patient's session (same and other hospital): each must answer "not found", change nothing and leave nothing in the owner's audit trail. Adding a route without a probe fails the test. These probes and the 15 scenarios found 8 defects, all fixed with a test that failed first (chapter 21).

19.6 Data map (summary)

Data	Where	Notes
Name, phone, ABHA id	`patients`	Encrypted; phone digest not unique (shared family phones)
OPD number / PIN	`patients.mrn` / not stored	PIN derived; keep the slip like a hospital card
Answers, transcripts	`interview_turns` ; Redis until confirmation	Encrypted, bound to the turn
Case sheet, AI draft	`case_sheets`	Encrypted JSON; only the repository decrypts
Old papers	`documents` , `document_contents`	Encrypted photo and reading; no-store; views audited
Consents, timeline, audit	`consents` , `visit_events` , `audit_log`	Append-only for the app
Offline kiosk forms	Browser IndexedDB	AES-GCM, deleted after confirmed sync
Phone check-in ticket	Phone local storage	Removed the first time the page opens on a later day

Not done yet (Phase 7)

HTTPS, MFA and password reset, key rotation, scheduled and approved retention, encrypted off-site backups, image scanning, DAST, API fuzzing and an independent penetration test; a legal DPDP review, processor agreements and AI provider contracts. `docs/PRIVACY.md` is an engineering inventory, not a legal opinion.

20 Offline use and the PWA

- **Installable app.** `vite-plugin-pwa` precaches the app shell and the recorded prompt audio. It updates itself: after a rebuild, an open tab moves to the new version on its next reload.
- **Secure-context safe.** Request ids use `crypto.randomUUID` where available and otherwise an RFC 4122 v4 id from `crypto.getRandomValues`, so every action works on plain-HTTP LAN pages (where the browser withholds `randomUUID`). The offline form is offered only where Web Crypto exists.
- **Retry-safe network.** Answers, corrections and confirmations can be re-sent safely; the kiosk says "tap again, your answers are kept"; the status page and the staff board reconnect by themselves.

Capability	No internet, LAN up	Server down	Phone off the network
Kiosk and phone interview (tap/type), tokens, live queue, staff console, case sheets, PDF	✓	✗	✗
Rule-based red flags, routing, triage score	✓	✗	✗
Browser speech recognition (Chrome)	✗ (uses Google's servers)	✗	✗
Sarvam/Gemini speech, AI drafts, paper reading	✗ (falls back to tap/type and the rule-based sheet)	✗	✗
Kiosk offline short form (encrypted outbox, replay on reconnect)	n/a	✓ on the kiosk only	✗

21 Quality: tests, CI and evaluations

965
API tests passed (2 skipped: the load smoke needs a running stack)

96%
coverage of `core` and `domain` (CI gate 95%)

96
web unit tests (Vitest)

144
browser tests (Playwright) with axe accessibility scans, plus a PWA offline suite

100
Lighthouse accessibility on kiosk and status pages (performance 89–93)

15/15
end-to-end scenarios covered; 8 defects found and fixed

21.1 What runs on every change (GitHub Actions)

Job	Steps
backend	Real PostgreSQL 16 and Redis services; database roles created; hash-locked install; ruff; `mypy --strict`; question-bank validation; prompt-audio manifest check; interview golden set; red-flag, extraction, adversarial and AI red-flag evaluations; evaluation outputs must be unchanged; case-sheet schema unchanged; migrations up → base → up; full pytest with coverage; 95% gate on core and domain.
frontend	ESLint; TypeScript; Vitest; Playwright e2e; PWA e2e; production build; Lighthouse CI budgets; locale parity; colour-contrast gate (WCAG AA); OpenAPI regenerated and the generated TypeScript types must be unchanged.
security	Gitleaks over the pushed or pull-request commits; Bandit; pip-audit for both lock files (one documented waiver, ADR 006); Semgrep (`--config auto`, fails on any finding); npm audit (high).

`main` is protected: changes arrive only through pull requests with all three jobs green.

21.2 Synthetic evaluations

Evaluation	Result
Red flags, 500 scripts (en 200, hi 150, mr 100, Hinglish 50): dev 350 / test 150, precision and recall	1.0 / 1.0
Must-catch red flags (test set)	60 / 60
Negations not falsely flagged ("no chest pain")	50 / 50 test, 88 / 88 dev
Department routing, top-1	224 / 224 dev, 96 / 96 test
Interview golden scenarios	30 / 30
History-extraction fixtures with zero invented fields	100 / 100
Adversarial prompts blocked by the output guard	100 / 100
Urgency never decreases when a warning sign is added	500 / 500

What these numbers do and do not mean

The cases are generated from the same phrase vocabulary as the rules, so they prove the software behaves as designed. They do not estimate real-world clinical performance; that needs clinician-labelled data from a pilot (Phase 7 task 7.6.5).

21.3 The fifteen end-to-end scenarios

#	Scenario	What must happen
1	First-time patient on a phone	Same interview as a kiosk over plain-HTTP Wi-Fi; token and live status; "Phone" badge

#	Scenario	What must happen
2	Returning patient	OPD card links today's visit (age/sex must fit); the doctor sees earlier visits; staff can link without the card
3	Incomplete interview	Never queued; idle reset ends it with no token; a tapped danger sign still alerts; a phone resumes
4	Red-flag symptom	Urgent at once; staff alert within a second (p95); "Nurse to confirm" until acknowledged
5	Duplicate submission, slow network	A double tap or retry never creates a second answer, visit or token
6	Called patient does not appear	Marked absent after the grace period; keeps token, urgency and arrival; status page says go to the desk
7	Skip and recall	Recall restores the original arrival; only a skipped patient can become a no-show
8	Patient returns after being marked absent	Recalled while open; after a no-show, registers again and the card joins the same record
9	AI provider unavailable	Check-in, triage, routing and token work; rule-based sheet; a hanging provider adds at most ~5 s
10	Network interruption	Re-sent answers saved once; "tap again"; offline form syncs once; boards and status pages reconnect
11	Two doctors in conflict	Different patients on simultaneous "Call next"; one patient per doctor
12	Another patient's records	A patient session reaches only its own visit; another department gets "not found"
13	Another hospital's records	Every record route answers "not found"; nothing changes; no audit entry in the owner's trail
14	Invalid or oversized document	Only real JPEG/PNG/WebP under 1 MB; refusals explained in the patient's language; per-visit limits
15	Simultaneous registrations	Distinct tokens in order with no gap; each patient once on the board

Defects the scenarios found (each now has a test that failed before the fix): a re-sent answer refused with an English error; a session able to cancel a queued visit; corrections after a lost confirmation reopening the interview; any file sent as a "photo" reaching the model; unlimited "Call staff"; answers left in Redis for two hours after confirmation; a deactivated nurse's board still receiving alerts; another hospital's session id answered 410 instead of 404.

21.4 Load

Run (Docker Desktop on a laptop, 2 API workers, AI off)	Result
40 patients arriving within 10 s, 4 staff boards refreshing every 2 s, every patient on their status page	Done in 14.1 s; 0 failed requests; every step p95 ≤ 107 ms (answers 105 ms, confirmation 86 ms)
Stress: 150 patients within 10 s	0 failed requests; every token unique and in order; answers p95 1.7–2.0 s; confirmation p95 3.0 s
5,000-patient queue in the browser	An urgent change applied in under 100 ms
Reports summary query, 100,000 visits	1.88 ms p95

A busy OPD sees well under one new patient per second; the smoke runs four a second. These are single-machine lab results, not a capacity guarantee: realistic multi-hour and multi-hospital tests are Phase 7 tasks 7.8.2–7.8.10.

PART III

Running it

How to start the demo, install OPD Saathi at a hospital, run it day to day, fix common problems, and what still stands between today's build and a real hospital deployment.

22. Running the demo
23. Installing at a hospital
24. Day-to-day operations and commands
25. Troubleshooting
26. Limitations and the road to production
27. Glossary

22 Running the demo

The demo is a complete fictional hospital started by one script on a Windows PC with Docker Desktop. It must never be used for a real hospital.

```
# from the repository root, with Docker Desktop running
powershell -ExecutionPolicy Bypass -File .\start-demo.ps1          # first run builds images (several
minutes)
powershell -ExecutionPolicy Bypass -File .\start-demo.ps1 -NoBuild # later runs, seconds
powershell -ExecutionPolicy Bypass -File .\scripts\demo-check.ps1  # is everything ready?
powershell -ExecutionPolicy Bypass -File .\scripts\lan-check.ps1   # can phones reach this PC?
```

What `start-demo.ps1` does

1. Checks Docker; creates `.env` with random secrets if missing (and fills in signing keys an older `.env` lacks, so the published development defaults are never used).
2. Picks the web port (5173, or 8080 if busy) and this PC's Wi-Fi/LAN address for the QR links (`PUBLIC_WEB_URL`), and opens the web port to the network.
3. Starts PostgreSQL and Redis, applies migrations, seeds the **Demo Government Hospital, Nagpur**: 8 departments, one doctor per department (`doctor01` … `doctor08`), five General Medicine nurses (`nurse01` … `nurse05`) and an administrator, all `@demo.opdsathi.test` .
4. Adds a believable month of finished OPD days for Reports, and resets the fictional returning patient (Sunita, 52, with an OPD card, an earlier visit three weeks ago and an old prescription photo kept unread).
5. Starts the API, worker and web app; creates a kiosk activation link and a phone check-in link; prints the staff logins and the returning patient's OPD number and PIN; opens the kiosk in the default browser.

> **Passwords**
>
> Demo passwords live only in `.env` (`DEMO_ADMIN_PASSWORD` , `DEMO_STAFF_PASSWORD`) and are printed by the script; they are deliberately not written in any document.

`demo-check.ps1` checks the services, readiness, the sign-in hospital list, the database migration level, which AI and voice providers were selected, a tiny live answer from Gemini and Sarvam (no patient data), the returning patient, and the synthetic papers in `docs/demo-papers/` . Run `start-demo.ps1` again after every reboot or network change: QR codes must point at the PC's current address.

The ten-minute story (summary of `docs/DEMO.md`)

Time	Screen	Do
0:00	Kiosk	Marathi → Start; care consent; **Yes, use AI help**; age 52, Female; OPD card number and PIN from `start-demo` ; "the same problem"
1:40	Kiosk	Safety screen: none of these; answer by tapping, speak one answer in Marathi
3:15	Kiosk	Read-back plays; under Old papers tap Test report and photograph `docs/demo-papers/synthetic-lab-report.png` ; confirm
4:00	Phone	Scan the token QR: live status and **Stop AI help**
4:30	Nurse	The queue board, live, urgency first

Time	Screen	Do
5:00	Doctor	Call, start; the cited AI brief; a source chip opens the transcript; Accept; Previous visits; Old papers with printed-range flags; Matches the photo; Finish; PDF
8:00	Admin	Reports; Settings → Departments (Ayurveda switch); Settings → Kiosks → Show QR
+1 min	Kiosk + nurse	Emergency: a new check-in taps Chest discomfort; the urgent alert appears at once; the patient goes first, "Nurse to confirm"

Before the next run-through: `docker compose run --rm api python -m app.cli seed-demo-returning --reset` removes visits joined to the returning patient and keeps her card.

23 Installing at a hospital

Each hospital gets its own installation: a PC or server inside the hospital network with Docker, its own database and its own keys. Read chapter 26 first: today's build is ready for supervised pilots with synthetic data, not for real patients.

```
cd C:\OPD-Saathi
Set-ExecutionPolicy -Scope Process -ExecutionPolicy Bypass
.\setup-hospital.ps1                # first installation; safe to run again (keeps keys and passwords)
.\setup-hospital.ps1 -NoBuild       # restart quickly after editing .env
.\setup-hospital.ps1 -AddHospital   # add another hospital on the same server
.\setup-hospital.ps1 -WebPort 8080  # use another port
```

The script creates `.env` with fresh random passwords and keys, builds and starts the stack, applies migrations, asks for the hospital's name, a short unique code and the first administrator, creates them with the 8 standard departments, and prints the addresses for staff PCs and kiosks. On Linux without PowerShell the same steps are done by hand:

```
cp .env.example .env    # replace every "replace-…" value:
                        #   passwords and JWT_SECRET: openssl rand -hex 32
                        #   APP_DATA_KEY, APP_HMAC_KEY, CURSOR_HMAC_KEY: openssl rand -base64 32
                        #   add WEB_BIND=0.0.0.0, WEB_ORIGIN and PUBLIC_WEB_URL=http://<server-ip>:5173
docker compose up -d postgres redis redis-cache worker-role-bootstrap
docker compose run --rm api alembic -c /app/alembic.ini upgrade head
docker compose up -d api worker web
docker compose run --rm api python -m app.cli setup-hospital
```

The administrator's first day

1. Sign in at `/staff` using the server's network address (not `localhost`), and open Settings.
2. **Departments:** keep, rename, deactivate or delete the standard ones; add others; set the call grace period; decide on the Ayurveda history.
3. **Users:** add every doctor and nurse with their department.
4. **Kiosks:** register each kiosk and open its setup link once on that device (Chrome or Edge, F11); create a phone check-in QR if patients will use their own phones.
5. Take the first backup and store `.env` safely with it (without `APP_DATA_KEY` the encrypted fields can never be read).

The `.env` file

Setting	What it is
`APP_DATA_KEY`, `APP_HMAC_KEY`	32-byte keys (base64) that encrypt patient fields and derive lookup digests, PINs, status links and check-in codes. Never change them after patients exist; back them up separately.
`JWT_SECRET`, `JWT_KID`, `CURSOR_HMAC_KEY`	Sign staff, kiosk and patient tokens and list cursors. Changing them signs everyone out (kiosks then need a new activation code).
`POSTGRES_PASSWORD`, `OSA_APP/MIGRATOR/WORKER_PASSWORD`, `REDIS_PASSWORD`	Database and cache passwords, one per role.
`APP_ENV`	`development`, `test` or `production` (production enforces strong keys and hides API docs).

Setting	What it is
`WEB_PORT`, `WEB_BIND`, `WEB_ORIGIN`, `PUBLIC_WEB_URL`	The address and port staff, kiosks and phones use; allowed browser origins; the address encoded in QR codes.
`GEMINI_API_KEY`, `SARVAM_API_KEY`, `LLM_PROVIDER`, `STT_PROVIDER`, `TTS_PROVIDER`	Optional AI and voice; "auto" uses whichever keys exist. `LLM_ENABLED=false` is the global AI kill switch; `PROVIDER_MODE=demo-safe` uses canned fixtures.
`LLM_MODEL`, `LLM_FALLBACK_MODEL`, `LLM_THINKING_LEVEL`, `LLM_MAX_CALLS_PER_SESSION`, `LLM_MAX_OUTPUT_TOKENS`, `AI_FOLLOWUP_QUESTIONS`	Model choice and limits (calls 1–30, tokens 1–1024).
`NOTIFICATION_PROVIDER`, `TWILIO_*`, `WHATSAPP_*`	Queue messages: `console` (default), Twilio SMS or WhatsApp.
`METRICS_ENABLED`, `GRAFANA_*`, `LOG_LEVEL`	Optional local metrics profile; log level.
`RETENTION_DAYS`, `ANONYMISE_DAYS`	Defaults (30, 90) for the retention command; not applied automatically.
`DEMO_*`	Demo script only.

Some keys in `.env.example` are not read by the code yet (`DATABASE_READ_URL`, `DB_POOL_*`, `WEB_CONCURRENCY`, `MAX_QUESTIONS`, `S3_*`, `SENTRY_DSN`, `OTEL_*`, `READBACK_LLM`); Phase 7 task 7.0.3 wires or removes them.

24 Day-to-day operations and commands

Administrators need none of these day to day: Settings covers staff, kiosks and departments. These are for whoever runs the server. `cli` below stands for `docker compose run --rm api python -m app.cli`.

Command	What it does
`docker compose ps` · `docker compose logs -f api`	Service health; follow the API log (worker, web likewise). Logs never contain patient data.
`docker compose down`	Stop everything; data is kept. **Never** `down -v` on a hospital server: it deletes the database.
`docker compose run --rm api alembic -c /app/alembic.ini upgrade head`	Apply database migrations (the scripts do this).
`cli setup-hospital` / `cli create-admin --hospital-id <id>`	Create a hospital with its first administrator and standard departments / add an administrator (recovery when the only admin is locked out).
`cli create-kiosk --hospital-id <id> --label "OPD entrance"`	Register a kiosk without Settings; prints a one-time code.
`cli queue-rebuild`	Rebuild the live queues from PostgreSQL (after Redis data loss).
`cli seed-demo`, `seed-demo-data` / `clear-demo-data`, `seed-demo-history --days 30` / `clear-demo-history`, `seed-demo-returning [--reset]` / `clear-demo-returning`, `demo-kiosk`, `demo-checkin`, `seed-bulk` / `clear-bulk`	Demo and load-test data (synthetic only).
`cli ai-check`	One tiny live request to Gemini and Sarvam (no patient data).
`sh scripts/backup.sh`	Database dump and Redis snapshot to `backups/<time>/` with checksums and a visit-count manifest, owner-only permissions. Store `.env` separately and safely.
`RESTORE_DATABASE=osa_restore_x sh scripts/restore.sh backups/<time>`	Restore into a separate database (refuses the live one), verify checksums and the visit count.
`make retention HOSPITAL_ID=<id>` / `make retention-apply …`	Dry run / apply retention: delete old interview turns, summaries, red-flag events and paper photos; pseudonymise patients whose visits are all old.
`.\scripts\lan-check.ps1 [-AddFirewallRule | -RemoveFirewallRule]`	Diagnose phone access: LAN address, network profile, bindings, firewall, and whether QR links match this PC.
`make load-smoke`, `make chaos`, `make scan`, `make sbom`, `make eval`	Load smoke on the running stack (own synthetic hospital); failure drills; security scans; SBOM; evaluations.

Updating to a new version

1. Take a backup and keep a copy of `.env`.
2. Replace the folder with the new version (or `git pull`), keeping `.env`.
3. Run `.\setup-hospital.ps1`: it rebuilds, migrates and restarts. Data, staff and kiosks are kept; kiosks stay signed in.
4. Open tabs pick up the new version on their next reload.

A typical morning

The server is already running; kiosks show "Start patient check-in"; yesterday's unfinished visits were closed automatically, so every queue starts empty. The administrator glances at Settings → Kiosks ("N of M active kiosks online

now"); an offline kiosk usually just needs its device switched on or reconnected.

25 Troubleshooting

You see	Why, and what to do
Kiosk shows the activation screen	The kiosk was deleted, set up on another device, or its browser data was cleared. Settings → Kiosks → ⋯ → New activation code → open the setup link on the kiosk.
"The activation code is invalid, expired, or has already been used"	Codes work once and for 24 hours. Issue a new code.
Setup link or QR opens a blank page on another device	It was made from a `localhost` page or the PC's address changed. Open Settings with the server's network address; run `start-demo.ps1` / `setup-hospital.ps1 -NoBuild` again after a network change.
Phones cannot open the QR	Run `lan-check.ps1`. Usual causes: the QR points at an old address; the Wi-Fi isolates devices (use a dedicated router or hotspot); Windows marks the network Public (set it Private, or add the narrow rule the script prints). Never turn the firewall off.
"Tap and speak" missing on a network kiosk or phone	Browsers allow the microphone only on HTTPS or on the server itself. Tapping and typing work; HTTPS is Phase 7 task 7.2.1. Managed kiosk PCs can start Chrome with `--unsafely-treat-insecure-origin-as-secure=http://<server>:5173`.
Kiosk: "This kiosk is turned off"	An administrator deactivated it: Settings → Kiosks → Activate.
Kiosk: "Ask staff to check this kiosk"	No active department: activate at least one (normally General Medicine).
Staff: "account locked"	Five wrong passwords lock an account for 15 minutes. Wait, or re-create the account; `create-admin` for a locked-out sole administrator.
"Department still in use" when deleting	Finish its open visits and move its staff first, or deactivate it instead.
"Already exists" when creating	That email, department code or token prefix is used in this hospital.
A paper reading stays "AI is reading this paper"	The reading failed or timed out (shown as failed after 2 minutes). The photo is the record: read it directly.
An old version after an update	Reload once; the app updates itself from then on.
Worker log repeats "relay_queue_outbox failed … APP_DATA_KEY"	The known notification defect (section 16.11, task 7.0.1). Queue updates still work; it appears once any patient opted in to SMS.
Nothing loads / "Cannot connect to the Docker daemon"	Start Docker Desktop, wait for "Engine running", then run the script again with `-NoBuild`.

26 Limitations and the road to production

Every feature in the IIT Bombay plan is built and tested, and the system runs reliably as a demonstration on synthetic data. It is **not** ready for real patients yet. The final phase, Phase 7, is planned in `docs/PHASE_7_PRODUCTION_PLAN.md`: 17 workstreams and 131 tasks, about 12 weeks for three engineers, with clinical and legal tracks in parallel.

What stands between today and a real hospital

Needs people outside the team

- Clinical sign-off of every question, rule, weight and Ayurveda item; native-speaker review of Hindi and Marathi.
- DPDP Act: patient notice, processor agreements, retention schedule, breach procedure; a regulatory (medical-device) assessment.
- Paid or enterprise AI contracts with data-processing terms, or AI switched off.
- Hospital IT: certificates, network segmentation, backup storage, a pilot site.

Engineering

- HTTPS; a production configuration that cannot fall back to development secrets.
- Password reset, MFA for administrators, key rotation and escrow.
- Encrypted, off-site, point-in-time backups with restore drills; monitoring that alerts a person.
- Image scanning, DAST, API fuzzing, an independent penetration test; real load and 8-hour soak tests.

Two defects found while preparing Phase 7

#	Defect	Task
F1	The worker's notification step fails every 2 seconds once a patient opts in to SMS: the worker has no data key but tries to decrypt the phone number (300 failures in 10 minutes on the demo stack). Queue updates are unaffected.	7.0.1 (P0)
F2	Typing a phone number ticks the SMS opt-in by itself; consent must be the patient's own tap.	7.0.2 (P0)

Features not built yet

Feature	Status
Doctor's own examination, assessment and plan notes	Planned (7.11.1)
Review-of-systems screen; papers timeline and prior investigations	Planned (7.11.2, 7.11.3)
Screens for staff editing, password reset, the AI switch, the audit log; lost-card re-issue	Planned (7.3, 7.11.4, 7.11.5)
Real SMS/WhatsApp (DLT registration in India); guardian answering for a patient	Planned (7.11.6, 7.11.7)
FHIR export; ABHA "Scan & Share" and ABDM milestones; hospital information system interface	Planned (7.10); ABDM certification is a months-long external process
Patients viewing their own records; drug-interaction checks; general reference ranges; RAG assistant	Deliberately out of v1.0 (stronger patient sign-in or a licensed knowledge base needed; RAG is design-only)
More languages	After the current three are reviewed; the add-a-language path exists

Phase 7 gates

Gate	Week	Passes when
G0 Fixed	1	F1 and F2 fixed with tests; unread settings removed or wired
G1 Secure build	4	Production overlay and smoke in CI; HTTPS on staging; password reset and admin MFA; scans clean; threat model
G2 Operable	7	Encrypted off-site point-in-time backups and restore tests; alerts reach a phone; runbooks; load and soak met
G3 Pilot-ready	10	Penetration test closed; clinical and language sign-off; legal documents signed; usability test; real-device HTTPS check; training
G4 Live	12+	Pilot thresholds met; go-live checklist signed; `v1.0.0` tagged

27 Glossary

Term	Meaning
OPD	Outpatient department: where patients see a doctor without being admitted.
Token	The patient's number for the day: department prefix and counter (`MED-016` = 16th General Medicine patient today).
OPD number / card / PIN	The hospital registration number (`26-000123`) and a six-digit PIN on the token slip; together they link a later visit to the same record.
Danger tile	One of six large buttons on the safety screen for signs that need immediate attention.
Red flag	A warning raised by a safety rule, a danger tile, or an AI second opinion, from what the patient reported. Not a diagnosis.
Triage score	0–100, staff-only; decides the head start in the queue. Routine < 30 ≤ priority < 70 ≤ urgent (or any urgent red flag).
Head start	Minutes taken off a patient's arrival time: 0.6 per point, at most 45, bounded by the level.
Nurse to confirm	Marker on a patient who tapped a danger sign, until a nurse acknowledges the alert after seeing them.
Grace period	How long after a call staff must wait before marking a patient absent (default 3 minutes).
Skipped / recall / put back / no-show	Marked absent after a call / returned to the queue on the original arrival time / a mistaken call undone / a skipped patient who left.
Case sheet	The organised English history the doctor reads, with the patient's own words and the source of every field. Versioned.
AI draft / brief	The AI-written summary for the doctor; every sentence cites the answers it rests on; accepted, edited or rejected by a doctor.
Read-back	The summary the kiosk reads to the patient for confirmation, built from fixed templates.
Transcript	The unchangeable record of every question and the patient's exact answer.
Pertinent negative	A recorded "no" to listed symptoms ("Denies vomiting, loose motions"), kept apart from "don't know".
Old papers	Photos of prescriptions, test reports and discharge papers the patient brings; read by AI into a draft only with consent.
Printed range	The reference range printed beside a result on the same paper; the only range results are compared with.
Consent purpose	`care` (required) or `ai` (optional, withdrawable); each recorded separately.
Phone check-in QR	A printed QR that lets patients answer the interview on their own phone.
Activation code / device secret	A one-time 24-hour code that pairs a kiosk; the long-lived secret it receives.
Row-level security (RLS)	A PostgreSQL feature that makes each hospital's rows invisible to every other hospital.
Outbox	Events written in the same transaction as a change and relayed by the worker, so live screens always catch up.
Idempotency key	A unique key on each staff action so a repeat is recognised and not done twice.
PWA	Progressive Web App: a website that installs like an app and keeps working offline for the short form.
Ayurveda terms	Ahara Shakti (appetite), Jarana Shakti (digestion), Mala (bowel), Nidra (sleep), Vyayama Shakti (exertion); Ashtavidha Pariksha (eight-fold examination: Nadi, Mutra, Mala, Jihva, Shabda, Sparsha, Drik, Akriti); Prakriti, Vikriti and other Dashavidha items.
ABHA / ABDM / FHIR	Ayushman Bharat Health Account (national health id) / Ayushman Bharat Digital Mission / the HL7 standard for exchanging health records.
DPDP Act	India's Digital Personal Data Protection Act, 2023.

A All 96 API routes

All paths are served under `/api` by nginx. "Public" marks routes outside the authentication gate.

Method	Path	Purpose
Auth, kiosks and operations		
GET	`/auth/hospitals`	Public: hospital names for the sign-in picker
POST	`/auth/login` · `/auth/refresh` · `/auth/logout`	Public: staff sign-in, renew, sign-out
POST	`/kiosks/activate` · `/kiosks/token`	Public: pair a kiosk with a one-time code; renew its token with the device secret
POST	`/kiosks/heartbeat`	The kiosk's "I am online" signal
GET	`/healthz` · `/readyz` · `/metrics`	Public: liveness, readiness (PostgreSQL, Redis), metrics (if enabled)
GET	`/audit`	Administrator audit log, cursor-paginated
Intake (patient session)		
POST	`/intake/sessions`	Start a kiosk check-in (visit in `registering`)
POST	`/intake/assisted-sessions`	A nurse starts a check-in for a patient
GET · POST	`/intake/checkin/{code}` · `…/sessions`	Public: phone check-in point lookup; start on a phone
POST	`/intake/sessions/{id}/consent` · `…/consent/ai`	Care consent; the separate AI answer
POST	`…/identity` · `…/opd-card` · `…/follow-up`	Age, sex, phone; link with the OPD card; mark a follow-up
GET · POST	`…/danger-tiles` · `…/danger-signs`	The six tiles; record tapped ones (urgent alert)
GET	`…/interview/next-question`	The current question and progress
POST · PATCH	`…/interview/answers`	Answer; correct an earlier answer
POST	`…/interview/back` · `…/language`	Undo the last answer; switch language
POST	`…/help` · `…/needs-assistance`	"Call staff" alert; idle-on-review help flag
GET	`…/readback-audio`	Read-back speech, in memory, never cached
POST	`…/confirm`	Issue the token once (repeat-safe)
DELETE	`/intake/sessions/{id}`	End a check-in that has no token
POST	`/intake/offline-submissions`	Replay one encrypted offline kiosk form
GET · POST · DELETE	`/intake/sessions/{id}/documents[/{document_id}]`	The patient's old-paper photos: list, add, remove before confirming
Speech		
GET	`/intake/capabilities`	Which speech and AI features this visit may use
POST	`…/transcribe` · `…/speak` · `…/prescription-photo`	Transcribe one answer; speak approved text; read medicine names (photo not stored)
WS	`/ws/intake/{visit_id}/audio`	Stream audio for live transcription (never stored)
Queue, alerts and status		
GET	`/queue`	The department queue, missed calls and grace period
POST	`/queue/call-next`	Call the next patient (server chooses; serialised)

Method	Path	Purpose
POST	`/queue/{id}/call|start|complete|skip|recall|return|no-show|cancel`	Queue transitions (department-checked, idempotent)
POST	`/queue/{id}/override` · `…/routing/confirm`	Change urgency or department with a reason; confirm a low-confidence route
POST	`/queue/{id}/link-patient`	Join a visit to a registered patient at the desk
GET	`/patients` · `/patients/lookup`	Visits register; find a patient by OPD number or phone
GET · POST	`/alerts` · `/alerts/{id}/ack`	Unacknowledged alerts; acknowledge one
GET	`/status/{token}` · `…/stream`	Public: a visit's status (cached 2 s); live stream
POST	`/status/{token}/ai-consent/withdraw`	Public: the patient stops AI help
GET	`/visits/{id}/queue-events`	The visit's queue timeline
WS	`/ws/staff/queue`	Live queue deltas and alerts for staff boards
Case sheets and documents		
GET · PATCH	`/visits/{id}/case-sheet`	Read; edit with If-Match (restricted JSON patch)
GET	`/visits/{id}/case-sheet.pdf` · `…/transcript` · `…/history`	PDF; transcript; earlier visits (under-care rule)
POST	`/visits/{id}/ai-summary/review` · `…/ai-consent/withdraw`	Accept, edit or reject the AI draft; staff withdraw AI consent
GET · POST	`/visits/{id}/documents`	Papers of this visit and earlier ones; staff add a photo
GET · POST	`/visits/{id}/documents/{document_id}/content` · `…/review`	The photo (audited, no-store); the doctor's decision on the reading
Administration and reports		
GET · POST · PATCH · DELETE	`/admin/{users|kiosks|departments}[/{id}]`	List, create, update, delete (archive) staff, kiosks and departments
POST	`/admin/kiosks/{id}/activation-code`	New one-time code for a kiosk
GET · POST	`/admin/kiosks/{id}/checkin-link`	Show or replace a phone check-in QR
GET · PATCH	`/admin/ai-settings` · `/admin/queue-settings` · `/admin/interview-settings`	Hospital AI switch; call grace period; Ayurveda module
GET	`/analytics/summary` · `/analytics/timeseries` · `/analytics/export.csv`	Report cards, daily trend, CSV
GET · PATCH	`/analytics/settings`	Minutes saved per visit

B The question bank

English labels shown; every question and option also exists in Hindi and Marathi (`services/api/app/domain/interview/bank/`). All content is a draft awaiting clinical and native-speaker review. The danger tiles: chest discomfort; trouble breathing; heavy bleeding; fainting or unable to stay awake; sudden weakness or difficulty speaking; a seizure happening now.

Module	Question	Options
core	What is the main problem that brought you here today?	Fever; Cough; Difficulty breathing; Stomach pain; Vomiting or loose motions; Headache; Body or joint pain; Skin rash or itching; Burning while passing urine; Weakness or tiredness; Injury or fall (+ own words)
	Since when do you have this problem?	Since today; 1 to 3 days; 4 to 7 days; 1 to 4 weeks; More than a month
	How bad is it right now? (0–10)	Scale
fever	How is the fever?	Very high fever; With shivering or chills; Comes and goes; Mostly at night; All the time
	Along with the fever, do you have any of these?	Body ache; Headache; Cold or runny nose; Rash on skin; Burning urine; Vomiting; None of these
respiratory	Tell us about the cough or breathing.	Dry cough; Cough with phlegm; Blood in phlegm; Breathless on walking; Wheezing sound
	Do any of these apply to you?	Someone at home has TB; Had asthma before; Smoking or chulha smoke; Losing weight; None of these
digestive	Do you have any of these stomach problems?	Vomiting; Loose motions; Constipation; Acidity or burning; Loss of appetite; Blood or black stool; None of these
	Do you have stomach pain? If yes, where?	Upper stomach; Around the navel; Lower right side; Lower left side; Whole stomach; No stomach pain
urinary	Which of these do you have when passing urine?	Burning or pain; Going very often; Blood in urine; Passing very little urine; Cannot hold urine; None of these
	Do you also have any of these?	Fever; Pain in the side of the back; Swelling of feet or face; None of these
pain	Where is the pain?	Head; Chest; Back or waist; Joints like knee or shoulder; Legs; Arms; Stomach; Throat
	What is the pain like?	All the time; Comes and goes; Sharp or stabbing; Burning; Spreads to another place; Worse on moving
skin	What do you notice on your skin?	Rash or red spots; Itching; Boils or pus; Patches or colour change; Wound not healing; Swelling
	Where on the body is it?	Face; Hands or arms; Legs or feet; Chest / back / belly; All over the body
mobility (injury)	How did it happen?	Fell down; Road accident; Hurt at work or while playing; Animal or dog bite
	Since it happened, do you have any of these?	Cannot walk properly; Cannot lift the arm; Cannot bend; Swelling; Bleeding or open wound; None of these
sleep_energy	Have you noticed any of these?	Tired all the time; Losing weight; Poor sleep; Dizziness; Sweating at night; Feeling low or worried; None of these
past_history	Do you have any long-term illness?	Diabetes (sugar); High BP; Heart disease; Asthma; TB (now or before); Thyroid; No long-term illness
	Have you ever had an operation?	Caesarean (C-section); Appendix; Hernia; Gallbladder or stone; Eye (cataract); Bone or joint; Another operation; Never had an operation; Don't know; Prefer not to say
	Are you pregnant now? (only where it can apply)	Yes; Not sure; No
medications	Do you take any medicines regularly? You can also say their names.	Medicine for sugar; Medicine for BP; Painkillers; Thyroid medicine; No regular medicines (+ prescription scan with AI consent)
allergies	Are you allergic to any medicine or food?	No allergy; I don't know; Yes - to a medicine; Yes - to a food

Module	Question	Options
family_history	Does anyone in your family have any of these?	Diabetes; High BP; Heart disease; TB; Cancer; None of these
personal_history	Do you use any of these?	Smoking (bidi or cigarette); Chewing tobacco or gutkha; Alcohol; None of these; Prefer not to say
ayurveda (optional)	How is your hunger?	Good and on time; Less than before; More than before; Changes from day to day
	How do you feel after eating?	Light and fine; Heavy or bloated; Sour belching or burning; Gas
	How are your bowel movements?	Daily and easy; Hard or constipated; Loose; Not regular
	How do you sleep?	Sound sleep; Takes long to fall asleep; Wake up often; Sleepy in the day
	Can you do your daily physical work?	Easily; Get tired quickly; Get breathless; Cannot do it
	What do you usually eat?	Vegetarian; Non-vegetarian; Eggs; Mostly spicy or fried; Meals at irregular times
general_context	Is there anything else you want the doctor to know?	No - that is all (+ own words)

OPD Saathi collects history for clinical staff. It does not diagnose or recommend treatment. All patients, staff and hospitals shown in this document are fictional.